Problem 1 (Fun with indistinguishability).

In this problem, you are allowed to use the computational assumptions taught in class. For instance, you are allowed to assume the existence of (t, ε)-OWF for some (t, ε).

(a) Let $\mathcal{A}$ be some algorithm that runs in time T. Show that if $X \approx_{t,\varepsilon} Y$ for some $t > T$, then $\mathcal{A}(X) \approx_{(t-T),\varepsilon} \mathcal{A}(Y)$.

(b) Following the previous question, show that $\mathcal{A}(X) \approx_{t,\varepsilon} \mathcal{A}(Y)$ is not necessarily true.

(c) Let X, Y, Z be three arbitrary distributions (= random variables). Show that $X \approx_{t,\varepsilon} Y$ does not necessarily imply $(X, Z) \approx_{t,\varepsilon} (Y, Z)$.

Answer (a). We prove this by contradiction. Suppose to the contrary that $\mathcal{A}(X) \not\approx_{(t-T),\varepsilon} \mathcal{A}(Y)$, which means that there exists a probabilistic distinguisher $D^{\mathcal{A}(X),\mathcal{A}(Y)}$ running in time $t - T$ such that

$$\left|\Pr\left[D^{\mathcal{A}(X),\mathcal{A}(Y)}(\mathcal{A}(X)) = 1\right] - \Pr\left[D^{\mathcal{A}(X),\mathcal{A}(Y)}(\mathcal{A}(Y)) = 1\right]\right| \geq \varepsilon.$$

From this, we construct a reduction $D^{X,Y}$ (a probabilistic distinguisher) that runs in time t and distinguishes between X and Y with advantage at least ε as follows:

$$D^{X,Y}(s) \triangleq D^{\mathcal{A}(X),\mathcal{A}(Y)}(\mathcal{A}(s)).$$

That is, given $D^{\mathcal{A}(X),\mathcal{A}(Y)}$, who distinguishes between $\mathcal{A}(X)$ and $\mathcal{A}(Y)$ well enough, our $D^{X,Y}$ first applies $\mathcal{A}$ to the input s and then asks $D^{\mathcal{A}(X),\mathcal{A}(Y)}$ from which distribution $\mathcal{A}(s)$ was sampled.

The run time of $D^{X,Y}$ is hence

$$\text{run time of } \mathcal{A} + \text{run time of } D^{\mathcal{A}(X),\mathcal{A}(Y)} \leq T + (t - T) = t,$$

meaning $D^{X,Y}$ runs in time t. Moreover, $D^{X,Y}$ indeed distinguishes between X and Y with an advantage at least ε since

$$\left|\Pr\left[D^{X,Y}(X) = 1\right] - \Pr\left[D^{X,Y}(Y) = 1\right]\right| = \left|\Pr\left[D^{\mathcal{A}(X),\mathcal{A}(Y)}(\mathcal{A}(X)) = 1\right] - \Pr\left[D^{\mathcal{A}(X),\mathcal{A}(Y)}(\mathcal{A}(Y)) = 1\right]\right| \geq \varepsilon$$

by the construction of our $D^{X,Y}$.

This implies by definition $X \not\approx_{t,\varepsilon} Y$, which contradicts the assumption. ⊛

Answer (b). We show this by presenting a counterexample. Assume that there exists a (t, ε)-PRG $G : \{0,1\}^n \to \{0,1\}^m$ for some t, ε, n, m. Let D be a probabilistic distinguisher of minimal (worst-case) run time such that it distinguishes $G(U_n)$ from U_m with advantage at least ε, i.e.,

$$|\Pr[D(G(U_n)) = 1] - \Pr[D(U_m) = 1]| \geq \varepsilon. \tag{1.1}$$

Let t_D denote the (worst-case) run time of D. Since G is a (t, ε)-PRG and thus $G(U_n) \approx_{t,\varepsilon} U_m$, by the definition of D, we must have $t_D > t$.

To construct $\mathcal{A}$, we split all the steps of D into two halves: D_1 and D_2. D_1 behaves identically to D for time $T - \delta$ and then outputs its entire internal state (e.g., variables and data structures used by D) to "export" the execution state of D, where δ denotes the time for D_1 to output its entire internal state and for D_2 to read and restore it; correspondingly, D_2 reads as input the internal state exported by D_1 to "restore" it and then continues the execution of D for time $t_D - (T - \delta)$, outputting what D outputs. Thus by the construction, D_1 and D_2 run in time T and $t_D - T + 2\delta$, respectively, and we also have

$$D = D_2 \circ D_1. \tag{1.2}$$

For the counterexample, take $X \triangleq G(U_n)$, $Y \triangleq U_m$, and $\mathcal{A} \triangleq D_1$, and suppose that $t_D - T + 2\delta \leq t$. (Note that D_1m runs in time T, as required for a counterexample.) Since $G(U_n) \approx_{t,\varepsilon} U_m$ is already given by the definition of PRG, it remains to show that $\mathcal{A}(G(U_n)) \not\approx_{t,\varepsilon} \mathcal{A}(U_m)$. To see this, observe that D_2 runs in time $t_D - T + 2\delta \leq t$, as we suppose, and distinguishes between $\mathcal{A}(G(U_n))$ and $\mathcal{A}(U_m)$ with advantage at least ε since

$$\begin{aligned}
&|\Pr[D_2(\mathcal{A}(G(U_n))) = 1] - \Pr[D_2(\mathcal{A}(U_m)) = 1]| \\
=&|\Pr[D_2(D_1(G(U_n))) = 1] - \Pr[D_2(D_1(U_m)) = 1]| \\
=&|\Pr[D(G(U_n)) = 1] - \Pr[D(U_m) = 1]| && \text{(by (1.2))} \\
\geq&\varepsilon. && \text{(by (1.1))}
\end{aligned}$$

This implies by definition that $\mathcal{A}(G(U_n))$ and $\mathcal{A}(U_m)$ is not (t,ε)-indistinguishable, as desired. ⊛

Answer (c). We show this by presenting a counterexample. Assume that there exists a (t,ε)-PRG $G : \{0,1\}^n \to \{0,1\}^m$ for some t, ε, n, m such that $\varepsilon \leq 1 - \frac{1}{2^m}$ and that t is no less than the run time of the distinguisher $D^{(X,Z),(Y,Z)}$, which is defined later.

As a counterexample, take $X \triangleq G(U_n)$ and $Y \triangleq Z \triangleq U_m$. Note that we regard distributions X, Y, Z as random variables over $\{0,1\}^m$ in this context. This means that when sampling a 3-tuple $(x,y,z) \leftarrow (X,Y,Z)$, we always have $x = G(u)$ for some $u \xleftarrow{\$} \{0,1\}^n$ and, most importantly, $y = z \xleftarrow{\$} \{0,1\}^m$.

Showing $X = G(U_n) \approx_{t,\varepsilon} U_m = Y$ is straightforward, as it is given directly by the assumption that G is a (t,ε)-PRG. On the other hand, to show $(X,Z) \not\approx_{t,\varepsilon} (Y,Z)$, we construct a probabilistic distinguisher $D^{(X,Z),(Y,Z)}$ that (1) runs in time t and (2) distinguishes between (X,Z) and (Y,Z) with advantage at least ε, as follows:

$$D^{(X,Z),(Y,Z)}(s_1,s_2) \triangleq [s_1 = s_2] = \begin{cases} 0, & \text{if } s_1 \neq s_2 \\ 1, & \text{if } s_1 = s_2 \end{cases}.$$

The first part is given directly by our assumption that t is no less than the run time of $D^{(X,Z),(Y,Z)}$. Just to clarify, it is reasonable to assume that $D^{(X,Z),(Y,Z)}$ runs in time t since the time complexity of $D^{(X,Z),(Y,Z)}$ is only $O(m)$. For the second part, we simply calculate that

$$\begin{aligned}
&\left|\Pr\left[D^{(X,Z),(Y,Z)}(X,Z) = 1\right] - \Pr\left[D^{(X,Z),(Y,Z)}(Y,Z) = 1\right]\right| \\
=&|\Pr[X = Z] - \Pr[Y = Z]| && \text{(by the construction of } D^{(X,Z),(Y,Z)}) \\
=&|\Pr[X = Z] - 1| && \text{(since we have assumed } Y = Z) \\
=&|\Pr[G(U_n) = U_m] - 1| && \text{(by definitions of } X \text{ and } Z) \\
=&\left|\frac{1}{2^m} - 1\right| && \text{(since } U_m \text{ is uniformly random)} \\
=&1 - \frac{1}{2^m} \\
\geq&\varepsilon, && \text{(by assumption)}
\end{aligned}$$

as desired. This implies by definition that (X,Z) and (Y,Z) is not (t,ε)-indistinguishable. ⊛

Problem 2 (Fun with PRF).

Assuming the existence of PRFs. Prove the statement "for all pseudorandom functions $F'_{k'} : \{0,1\}^n \to \{0,1\}^n$ with $k' \in \{0,1\}^n$ as the key, F_k is a pseudorandom function", or provide a counterexample to demonstrate that it is not true.

(a) $F_k(x) = F'_{k_1}(F'_{k_2}(x))$, where $k = (k_1, k_2)$. (Hint: Hybrid argument)

(b) $F_k(x) = F'_k(x) \oplus x$.

(c) $F_k(x) = F'_k(x) \oplus k$.

Answer (a). We prove F is a PRF by contradiction with hybrid arguments. Suppose to the contrary that F is not a PRF. As F' runs in deterministic polynomial time and so does F, it must be that there exists a PPT oracle distinguisher D such that

$$\varepsilon(n) \triangleq \left| \Pr_{k_1,k_2 \xleftarrow{\$} \{0,1\}^n} \left[D^{F'_{k_1}(F'_{k_2}(\cdot))}(1^n) = 1 \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[D^H(1^n) = 1 \right] \right| \tag{2.1}$$

is nonnegligible, where Func_n denotes the set of all functions mapping n-bit strings to n-bit strings.

Define randomized algorithms $\mathcal{H}_i$ for $i = 0, 1, 2$ that, given an oracle $\mathcal{O}$, output an oracle $\mathcal{H}_i(\mathcal{O})$:

$$\begin{aligned} \mathcal{H}_i(\mathcal{O}) &\triangleq F'_{U_n^{(1)}} \circ \cdots \circ F'_{U_n^{(i)}} \circ \mathcal{O} \circ U_{\mathsf{Func}_n}^{(i+2)} \circ \cdots \circ U_{\mathsf{Func}_n}^{(2)} \\ &\equiv F'_{k_1} \circ \cdots \circ F'_{k_i} \circ \mathcal{O} \circ H_{i+2} \circ \cdots \circ H_2 \quad \text{where } k_1, \dots, k_i \xleftarrow{\$} \{0,1\}^n \text{ and } H_{i+2}, \dots, H_2 \xleftarrow{\$} \mathsf{Func}_n. \end{aligned}$$

(Intuitively, $\mathcal{H}_i$ extends the oracle $\mathcal{O}$ to a composition of two functions, with $\mathcal{O}$ being the ith one.) We also define "hybrid distributions" Hyb_i for $i = 0, 1, 2$ using $\mathcal{H}$:

$$\mathsf{Hyb}_i \triangleq \mathcal{H}_i(U_{\mathsf{Func}_n}) \equiv \mathcal{H}(H) \quad \text{where } H \xleftarrow{\$} \mathsf{Func}_n. \tag{2.2}$$

Observe that for $i = 0, 1$,

$$\mathsf{Hyb}_{i+1} = \mathcal{H}_i(F'_{U_n}). \tag{2.3}$$

Moreover, note that $\mathsf{Hyb}_0 = U_{\mathsf{Func}_n}^{(1)} \circ U_{\mathsf{Func}_n}^{(2)}$ corresponds to composing two functions uniformly chosen from Func_n and is thus equivalent to the distribution U_{Func_n}, which is the same as that of H in (2.1); likewise, $\mathsf{Hyb}_2 = F'_{U_n^{(1)}} \circ F'_{U_n^{(2)}}$ is equivalent to the distribution of $F'_{k_1}(F'_{k_2}(\cdot))$ in (2.1).

From this, we construct a PPT oracle distinguisher (a reduction) $\mathcal{R}$, which distinguishes oracle access to F'_k (for uniform $k \xleftarrow{\$} \{0,1\}^n$) from oracle access to a uniform $H \xleftarrow{\$} \mathsf{Func}_n$ with nonnegligible advantage, by exploiting D as follows: given an oracle $\mathcal{O}$ (along with 1^n), $\mathcal{R}$ first samples $i \xleftarrow{\$} \{0,1\}$, passes a modified oracle $\mathcal{H}_i(\mathcal{O})$ to D (along with 1^n), and finally outputs what $D^{\mathcal{H}_i(\mathcal{O})}(1^n)$ outputs.

Clearly $\mathcal{R}$ runs in PPT. To see that $\mathcal{R}$ distinguishes oracle accesses to F'_{U_n} and U_{Func_n} with nonnegligible advantage, we calculate

$$\begin{aligned}
&\left|\Pr_{k \xleftarrow{\$} \{0,1\}^n}\left[\mathcal{R}^{F'_k}(1^n)=1\right]-\Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[\mathcal{R}^{H}(1^n)=1\right]\right| \\
=&\left|\Pr_{k \xleftarrow{\$} \{0,1\}^n, i \xleftarrow{\$} \{0,1\}}\left[D^{\mathcal{H}_i(F'_k)}(1^n)=1\right]-\Pr_{H \xleftarrow{\$} \mathsf{Func}_n, i \xleftarrow{\$} \{0,1\}}\left[D^{\mathcal{H}_i(H)}(1^n)=1\right]\right| && \text{(by the construction of } \mathcal{R}\text{)} \\
=&\left|\Pr_{i \xleftarrow{\$} \{0,1\}}\left[D^{\mathsf{Hyb}_{i+1}}(1^n)=1\right]-\Pr_{i \xleftarrow{\$} \{0,1\}}\left[D^{\mathsf{Hyb}_{i}}(1^n)=1\right]\right| && \text{(by (2.2) and (2.3))} \\
=&\left|\sum_{i=0}^{1}\frac{1}{2}\Pr\left[D^{\mathsf{Hyb}_{i+1}}(1^n)=1\right]-\sum_{i=0}^{1}\frac{1}{2}\Pr\left[D^{\mathsf{Hyb}_{i}}(1^n)=1\right]\right| \\
=&\frac{1}{2}\left|\sum_{i=0}^{1}\left(\Pr\left[D^{\mathsf{Hyb}_{i+1}}(1^n)=1\right]-\Pr\left[D^{\mathsf{Hyb}_{i}}(1^n)=1\right]\right)\right| \\
=&\frac{1}{2}\left|\Pr\left[D^{\mathsf{Hyb}_{2}}(1^n)=1\right]-\Pr\left[D^{\mathsf{Hyb}_{0}}(1^n)=1\right]\right| \\
=&\frac{1}{2}\left|\Pr_{k_1,k_2 \xleftarrow{\$} \{0,1\}^n}\left[D^{F'_{k_1}(F'_{k_2}(\cdot))}(1^n)=1\right]-\Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[D^{H}(1^n)=1\right]\right| && \text{(by the observation we made above)} \\
=&\frac{\varepsilon(n)}{2},
\end{aligned}$$

which is nonnegligible by assumption.

Hence, the existence of the PPT oracle distinguisher $\mathcal{R}$ implies F' is not a PRF, which is a contradiction. This completes our proof. ⊛

Answer (b). We prove F is a PRF by contradiction. Suppose to the contrary that F is not a PRF. As F' runs in deterministic polynomial time and so does F, it must be that there exists a PPT oracle distinguisher D such that

$$\begin{aligned}
\varepsilon(n) &\triangleq \left|\Pr_{k \xleftarrow{\$} \{0,1\}^n}\left[D^{F_k}(1^n)=1\right]-\Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[D^{H}(1^n)=1\right]\right| \\
&=\left|\Pr_{k \xleftarrow{\$} \{0,1\}^n}\left[D^{x \mapsto F'_k(x)\oplus x}(1^n)=1\right]-\Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[D^{H}(1^n)=1\right]\right| \qquad (2.4)
\end{aligned}$$

is nonnegligible, where Func_n denotes the set of all functions mapping n-bit strings to n-bit strings.

> Intuition. In the reduction, the distinguisher (as a subroutine) will give us x for each query. Knowing x, the reduction can thereby generate $F_k(x)$ from $F'_k(x)$, whose value can be obtained though querying the challenger.

From this, we construct a PPT oracle distinguisher (a reduction) $\mathcal{R}$, which distinguishes oracle access to F'_k (for uniform $k \xleftarrow{\$} \{0,1\}^n$) from oracle access to a uniform $H \xleftarrow{\$} \mathsf{Func}_n$ with nonnegligible advantage, by exploiting D as follows: On input 1^n, $\mathcal{R}$ runs $D(1^n)$. Whenever D queries $\mathcal{R}$ on x_i, $\mathcal{R}$ in turn queries the challenger on x_i, receives $y_i = \mathcal{O}(x_i)$ from the challenger, and then answer $y_i \oplus x_i = \mathcal{O}(x_i) \oplus x_i$ to D. Finally, when execution of D is done, output the bit returned by D. Equivalently, $\mathcal{R}$ wins the following distinguishing game G of oracle accesses to F'_{U_n} and U_{Func_n} with nonnegligible advantage:

$$
\begin{array}{ccccc}
\textsf{Ch} & & \mathcal{R} & & D \\
b \xleftarrow{\$} \{0,1\} & & & & \\
k \xleftarrow{\$} \{0,1\}^n,\ H \xleftarrow{\$} \mathsf{Func}_n & & & & \\
\mathcal{O} := \begin{cases} F'_k, & b = 0 \\ H, & b = 1 \end{cases} & & & & \\
 & \xrightarrow{\quad 1^n \quad} & & \xrightarrow{\quad 1^n \quad} & \\
 & \xleftarrow{\quad x_1 \quad} & & \xleftarrow{\quad x_1 \quad} & \\
 & \xrightarrow{\quad \mathcal{O}(x_1) \quad} & & \xrightarrow{\quad \mathcal{O}(x_1) \oplus x_1 \quad} & \\
 & \vdots & & \vdots & \\
 & \xleftarrow{\quad x_q \quad} & & \xleftarrow{\quad x_q \quad} & \\
 & \xrightarrow{\quad \mathcal{O}(x_q) \quad} & & \xrightarrow{\quad \mathcal{O}(x_q) \oplus x_q \quad} & \\
 & \xleftarrow{\quad b' \quad} & & \xleftarrow{\quad b' \quad} & \\
 & \mathcal{R} \text{ wins if } b' = b & & &
\end{array}
$$

Figure 1: Game G

Clearly $\mathcal{R}$ runs in PPT. To see $\mathcal{R}$ wins G with nonnegligible advantage, observe that:

- If $\mathcal{O} := F'_k$ for uniform $k \xleftarrow{\$} \{0,1\}^n$, then the answers it gives to D are distributed exactly as if D were interacting with an oracle $x \mapsto F'_k(x) \oplus x$ where $k \xleftarrow{\$} \{0,1\}^n$.
- If $\mathcal{O} := H$ for H chosen uniformly from Func_n, then
  1. for any query $x \in \{0,1\}^n$, since H is uniformly chosen from Func_n and therefore $H(x)$ is uniform, the answer $H(x) \oplus x$ is also uniform;
  2. all values of the function $x \mapsto H(x) \oplus x$ (which is a random variable), namely $\{H(x) \oplus x\}_{x \in \{0,1\}^n}$, are mutually independent since H is uniformly chosen from Func_n.

  This implies the function $x \mapsto H(x) \oplus x$ is uniformly distributed over Func_n, meaning that the answers $\mathcal{R}$ gives to D are distributed exactly as if D were interacting with an oracle uniformly chosen from Func_n.

In other words, given the oracle $\mathcal{O}$, the oracle $x \mapsto \mathcal{O}(x) \oplus x$ that $\mathcal{R}$ gives to D is distributed identically to what D expects. Using the observation above, we therefore calculate the advantage of $\mathcal{R}$ in distinguishing between the two oracle accesses as follows.

$$
\begin{aligned}
& \left| \Pr_{k \xleftarrow{\$} \{0,1\}^n} \left[\mathcal{R}^{F'_k}(1^n) = 1 \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[\mathcal{R}^{H}(1^n) = 1 \right] \right| \\
= & \left| \Pr_{k \xleftarrow{\$} \{0,1\}^n} \left[D^{x \mapsto F'_k(x) \oplus x}(1^n) = 1 \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[D^{x \mapsto H(x) \oplus x}(1^n) = 1 \right] \right| && \text{(by the construction of } \mathcal{R}) \\
= & \left| \Pr_{k \xleftarrow{\$} \{0,1\}^n} \left[D^{x \mapsto F'_k(x) \oplus x}(1^n) = 1 \right] - \Pr_{H' \xleftarrow{\$} \mathsf{Func}_n} \left[D^{H'}(1^n) = 1 \right] \right| && \text{(by the observation above)} \\
= & \varepsilon(n), && \text{(by (2.4))}
\end{aligned}
$$

which is nonnegligible by assumption.

Hence, the existence of the PPT oracle distinguisher $\mathcal{R}$ implies F' is not a PRF, which is a contradiction. This completes our proof. ⊛

Answer (c). F is not a PRF. We show this by giving a counterexample.
Assume that there exists a PRF $f_k : \{0,1\}^n \to \{0,1\}^n$ with key $k \in \{0,1\}^{n-1}$. We show that (1) the following keyed function $F'_k : \{0,1\}^n \to \{0,1\}^n$ with key $k \in \{0,1\}^n$ is a PRF whereas (2) $F_k : \{0,1\}^n \to \{0,1\}^n$ with key $k \in \{0,1\}^n$, defined by $F_k(x) \triangleq F'_k(x) \oplus k$, is not.

$$F'_{k_1 \| k_2}(x) \triangleq \begin{cases} f_{k_1}(x)_{[1:n-1]} \parallel k_2 & \text{if } x = 0^n \\ f_{k_1}(x) & \text{if } x \neq 0^n \end{cases} \quad \text{for all } k_1 \in \{0,1\}^{n-1}, k_2 \in \{0,1\}.$$

Intuition. This construction is similar to that of PRG $G(x) \triangleq G'(x_{[1:\frac{n}{2}]}) \parallel x_{[\frac{n}{2}+1:n]}$, given a PRG G'. Both constructions expose part of their randomness source (key of PRF and input of PRG) in output, as the definitions of PRG and PRF allow key/input to appear in output as long as the outputs as a whole are pseudorandom (nearly uniformly random).

Note that for $F'_{k_1 \| k_2}$ to become a PRF, we cannot embed part of the key, k_2, in more than one outputs (i.e., $F'_{k_1 \| k_2}(x) = f_{k_1}(x)_{[1:n-1]} \parallel k_2$ for multiple $x \in \{0,1\}^n$), since otherwise the outputs as a whole would not be pseudorandom (i.e. indistinguishable from uniformly random strings).

Claim 2.1. $F'_k : \{0,1\}^n \to \{0,1\}^n$ with key $k \in \{0,1\}^n$ is a PRF.

Proof. Suppose to the contrary that F' is not a PRF. As f runs in deterministic polynomial time and so does F', it must be that there exists a PPT oracle distinguisher D such that

$$\varepsilon(n) \triangleq \left| \Pr_{k \xleftarrow{\$} \{0,1\}^n} \left[D^{F'_k}(1^n) = 1 \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[D^H(1^n) = 1 \right] \right| \tag{2.5}$$

is nonnegligible, where Func_n denotes the set of all functions mapping n-bit strings to n-bit strings.

From this, we construct a PPT oracle distinguisher $\mathcal{R}$, which distinguishes oracle access to f_{k_1} (for uniform $k_1 \xleftarrow{\$} \{0,1\}^{n-1}$) from oracle access to a uniform $H \xleftarrow{\$} \mathsf{Func}_n$ with nonnegligible advantage for all $n \geq 2$, as follows:

On input 1^n, sample the latter part of key $k_2 \xleftarrow{\$} \{0,1\}$ and then run $D(1^n)$. Whenever receiving a query x_i from D, query the challenger on x_i, receive $\mathcal{O}(x_i)$ from the challenger, and then answer to D $\mathcal{O}(x_i)_{[1:n-1]} \parallel k_2$ if $x_i = 0^n$ and $\mathcal{O}(x_i)$ otherwise. Finally, when receiving the answer b' from D, output b' back to the challenger.

■

cont'd. Equivalently, $\mathcal{R}$ wins the following PRF game of f with nonnegligible advantage for all $n \geq 2$:

$$
\begin{array}{ccccc}
\textsf{Ch} & & \mathcal{R} & & D \\
b \xleftarrow{\$} \{0,1\} & & & & \\
k_1 \xleftarrow{\$} \{0,1\}^{n-1},\ H \xleftarrow{\$} \mathsf{Func}_n & & & & \\
\mathcal{O} := \begin{cases} f_{k_1}, & b = 0 \\ H, & b = 1 \end{cases} & \xrightarrow{\quad 1^n \quad} & k_2 \xleftarrow{\$} \{0,1\} & & \\
& & & \xrightarrow{\quad 1^n \quad} & \\
& \xleftarrow{\quad x_1 \quad} & & \xleftarrow{\quad x_1 \quad} & \\
& \xrightarrow{\quad \mathcal{O}(x_1) \quad} & y_1 := \begin{cases} \mathcal{O}(x_1)_{[1:n-1]} \parallel k_2 & \text{if } x_1 = 0^n \\ \mathcal{O}(x_1) & \text{if } x_1 \neq 0^n \end{cases} & \xrightarrow{\quad y_1 \quad} & \\
& \vdots & & \vdots & \\
& \xleftarrow{\quad x_q \quad} & & \xleftarrow{\quad x_q \quad} & \\
& \xrightarrow{\quad \mathcal{O}(x_q) \quad} & y_q := \begin{cases} \mathcal{O}(x_q)_{[1:n-1]} \parallel k_2 & \text{if } x_q = 0^n \\ \mathcal{O}(x_q) & \text{if } x_q \neq 0^n \end{cases} & \xrightarrow{\quad y_q \quad} & \\
& \xleftarrow{\quad b' \quad} & & \xleftarrow{\quad b' \quad} & \\
& & \mathcal{R} \text{ wins if } b' = b & &
\end{array}
$$

Figure 2: PRF game of f

Clearly $\mathcal{R}$ runs in PPT. To see that, for all $n \geq 2$, $\mathcal{R}$ distinguishes oracle accesses to f_{k_1} (for $k_1 \xleftarrow{\$} \{0,1\}^{n-1}$) and $H \xleftarrow{\$} \mathsf{Func}_n$ with nonnegligible advantage, observe that:

- If $\mathcal{O} := f_{k_1}$ for uniform $k_1 \xleftarrow{\$} \{0,1\}^{n-1}$, then on each query $x_i \in \{0,1\}^n$, the answer $\mathcal{R}$ gives to D is exactly $F'_{k_1\|k_2}(x_i)$, where $k_2 \xleftarrow{\$} \{0,1\}$ is chosen by $\mathcal{R}$ and thus fixed throughout the game. This means all answers that D receives are distributed exactly as if D were interacting with an oracle $F'_{k_1\|k_2}$ for some uniform $k_1 \xleftarrow{\$} \{0,1\}^{n-1}$ and $k_2 \xleftarrow{\$} \{0,1\}$.
- If $\mathcal{O} := H$ for H chosen uniformly from Func_n, then on each query $x_i \in \{0,1\}^n$ (including 0^n), since $H \xleftarrow{\$} \mathsf{Func}_n$ and $k_2 \xleftarrow{\$} \{0,1\}$ are uniformly and independently chosen, all possible answers, namely $\{H(x) \mid x \in \{0,1\}^n \setminus \{0\}\} \cup \{H(0^n)_{[1:n-1]} \parallel k_2\}$, are uniformly random and mutually independent. Notice that $H(0^n)_{[1:n-1]} \parallel k_2$ is also independent of all the other possible answers. This implies all answers that D receives are distributed exactly as if D were interacting with an oracle uniformly chosen from Func_n.

Putting both together, for all $n \geq 2$, the advantage of $\mathcal{R}$ is thus

$$
\begin{aligned}
&\left| \Pr_{k_1 \xleftarrow{\$} \{0,1\}^{n-1}} \left[\mathcal{R}^{f_{k_1}}(1^n) = 1\right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[\mathcal{R}^{H}(1^n) = 1\right] \right| \\
=&\left| \Pr_{k_1 \xleftarrow{\$} \{0,1\}^{n-1}, k_2 \xleftarrow{\$} \{0,1\}} \left[D^{F'_{k_1\|k_2}}(1^n) = 1\right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[D^{H}(1^n) = 1\right] \right| \\
=&\left| \Pr_{k \xleftarrow{\$} \{0,1\}^{n}} \left[D^{F'_{k}}(1^n) = 1\right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[D^{H}(1^n) = 1\right] \right| = \varepsilon(n), \qquad \text{(by (2.5))}
\end{aligned}
$$

which is nonnegligible by assumption. This breaks the pseudorandomness of f, implying that f is not a PRF, which is a contradiction. Hence F' must be a PRF, as claimed. ■

Claim 2.2. $F_k : \{0,1\}^n \to \{0,1\}^n$ with key $k \in \{0,1\}^n$, defined by $F_k(x) \triangleq F'_k(x) \oplus k$, is not a PRF.

Proof. To show this, we simply construct a naive PPT oracle distinguisher D that breaks the pseudorandomness of F, as shown below.

> On input 1^n, make a query 0^n to the challenger and receive as answer $\mathcal{O}(0^n)$. If the last bit of $\mathcal{O}(0^n)$ is 0, then output $b' = 0$; otherwise, output $b' = 1$.

Observation 2.3. For any $k = k_1 \parallel k_2 \in \{0,1\}^n$ (where $k_1 \in \{0,1\}^{n-1}, k_2 \in \{0,1\}$),

$$F_k(0^n) = F_{k_1\|k_2}(0^n) \triangleq F'_{k_1\|k_2}(0^n)\oplus(k_1\|k_2) \triangleq (f_{k_1}(0^n)_{[1:n-1]}\|k_2)\oplus(k_1\|k_2) = (f_{k_1}(0^n)_{[1:n-1]}\oplus k_1)\|0.$$

We show that D wins the following PRF game $\mathsf{G}_{\mathrm{PRF}}$ of F with nonnegligible advantage for all $n \geq 2$:

$$\begin{array}{ccc}
\mathsf{Ch} & & D \\
b \xleftarrow{\$} \{0,1\} & & \\
k \xleftarrow{\$} \{0,1\}^n,\ H \xleftarrow{\$} \mathsf{Func}_n & & \\
\mathcal{O} := \begin{cases} F_k, & b = 0 \\ H, & b = 1 \end{cases} & \xrightarrow{\quad 1^n \quad} & \\
 & \xleftarrow{\quad 0^n \quad} & \\
 & \xrightarrow{\quad \mathcal{O}(0^n) \quad} & \\
 & & b' := \begin{cases} 0 & \text{if } \mathcal{O}(0^n)_{[n]} = 0 \\ 1 & \text{otherwise} \end{cases} \\
 & \xleftarrow{\quad b' \quad} & \\
 & D \text{ wins if } b' = b &
\end{array}$$

Figure 3: Game $\mathsf{G}_{\mathrm{PRF}}$

Clearly D runs in PPT. Having Observation 2.3, we know the win rate of D in $\mathsf{G}_{\mathrm{PRF}}$ for all $n \geq 2$ is:

$$\begin{aligned}
&\Pr_{b\xleftarrow{\$}\{0,1\},k\xleftarrow{\$}\{0,1\}^n,H\xleftarrow{\$}\mathsf{Func}_n}[D \text{ wins } \mathsf{G}_{\mathrm{PRF}}] \\
=&\underbrace{\Pr_{b\xleftarrow{\$}\{0,1\}}[b=0]}_{=\frac{1}{2}}\underbrace{\Pr_{k\xleftarrow{\$}\{0,1\}^n}\left[F_k(0^n)_{[n]}=0\right]}_{=1\ (\text{by Observation 2.3})}+\underbrace{\Pr_{b\xleftarrow{\$}\{0,1\}}[b=1]}_{=\frac{1}{2}}\underbrace{\Pr_{H\xleftarrow{\$}\mathsf{Func}_n}\left[H(0^n)_{[n]}=1\right]}_{=\frac{1}{2}\ (\because H(0^n)\in\{0,1\}^n \text{ is uniformly random})} \\
=&\frac{1}{2}+\frac{1}{4},
\end{aligned}$$

where the advantage $\frac{1}{4}$ is nonnegligible.

This breaks the pseudorandomness of F, so F is not a PRF, as claimed. ∎

These two claims together complete our proof. ⊛

Problem 3 (Everything is One-Way function).

What if, for instance, OWF is in fact a much stronger primitive comparing to PRG? In this problem, you will be asked to fill up the gap by showing that the inverse direction of these statements are both easy and direct. In other words, prove the following statements with simple constructions: (For simplicity, in problem (a) and (c) you can assume the existence of a PRF $f : \{0,1\}^{l(n)} \times \{0,1\}^n \to \{0,1\}^n$, where $l(n) = \Theta(n^c)$ for some constant $c > 0$.)

(a) PRF implies OWF;

(b) PRG implies OWF;

(c) PRF implies PRG.

Answer (a). Assume that there exists a PRF $f : \{0,1\}^{l(n)} \times \{0,1\}^n \to \{0,1\}^n$ where $l(n) = \Theta(n^c)$ for some constant $c > 0$. Define an "inverse" function l' of l by

$$l'(m) \triangleq \min\{n \in \mathbb{N} \mid l(n) = m\}.$$

Intuition. Our goal is to define a OWF $g_m : \{0,1\}^m \to \{0,1\}^{p(m)}$ where $p(m) \gg m$. We want the codomain of g_m to be much larger than its image because values computed from f, which lie in the image of g_m (size $\leq 2^m$), will be easily distinguishable from values uniformly sampled from the codomain $\{0,1\}^{p(m)}$ of g_m (size $= 2^{p(m)} \gg 2^m \geq$ size of image) in the PRF game.

We show that the function $g_m : \{0,1\}^m \to \{0,1\}^{l'(m)p(l'(m))}$ defined by

$$g_m(k) \triangleq (f_k(0_2), \ldots, f_k((p(l'(m)) - 1)_2)),$$

where $(\cdot)_2$ denotes the $l'(m)$-bit representation of the given number and $p(n) = O(\mathsf{poly}(n))$ will be concretely specified later, is OW. Note that we require $p(n) = O(\mathsf{poly}(n))$ so that g_m is polynomial-time computable, as we will see later.

To show this, there are two properties to prove: (1) g_m is polynomial-time computable, and (2) g_m can be inverted with only negligible probability by any PPT adversary.

To see (1) holds, note that since $p(n) = O(\mathsf{poly}(n))$ and $f_k : \{0,1\}^n \to \{0,1\}^n$ can be computed in $\mathsf{poly}(n)$ time (as f is a PRF) where $n = l'(m)$, if $l'(m)$ is a polynomial of m, then g_m can be computed in $\mathsf{poly}(l'(m))p(l'(m)) = O(\mathsf{poly}(m))$ time. Therefore, it suffices to show that $l'(m) = \Theta(m^{c'})$ for some $c' > 0$. Specifically, we show that this indeed holds for $c' = \frac{1}{c} > 0$.

Claim 3.1. $l'(m) = \Theta(m^{c'})$ where $c' = \frac{1}{c} > 0$.

Proof. Since $l(n) = \Theta(n^c)$, there exist $a, b > 0$ and $N \in \mathbb{N}$ such that $an^c \leq l(n) \leq bn^c$ for all $n \geq N$. Observe that

$$m = l(l'(m)) \leq b(l'(m))^c \implies l'(m) \geq b^{-\frac{1}{c}} m^{\frac{1}{c}}$$

for all $m \geq max_{n<N} l(n) + 1$ and that

$$m = l(l'(m)) \geq a(l'(m))^c \implies l'(m) \leq a^{-\frac{1}{c}} m^{\frac{1}{c}}$$

for all $m \geq \max_{n<N} l(n)+1$. Combining the two, we conclude that for all $m \geq \max_{n<N} l(n)+1$,

$$b^{-\frac{1}{c}} m^{\frac{1}{c}} \leq l'(m) \leq a^{-\frac{1}{c}} m^{\frac{1}{c}},$$

which implies $l'(m) = \Theta(n^{c'})$ where $c' = \frac{1}{c} > 0$ as $c > 0$. ∎

To see (2) holds, we prove it by contradiction. Suppose to the contrary that there exists a PPT adversary $\mathbf{Adv}^{OW}$ that wins the following inverting game of g_m, denoted G, with probability $\varepsilon(m)$ for all $m \in \mathbb{N}$, where ε is nonnegligible.

$$\begin{array}{ccc}
\textsf{Ch} & & \mathbf{Adv}^{OW} \\
k \leftarrow \{0,1\}^m & & \\
 & \xrightarrow{(1^m, y := g_m(k))} & \\
 & \xleftarrow{\quad k' \quad} & \\
\multicolumn{3}{c}{\mathbf{Adv}^{OW} \text{ wins if } g_m(k') = g_m(k)}
\end{array}$$

Figure 4: Game G

By abusing $\mathbf{Adv}^{OW}$, we construct as follows a PPT oracle distinguisher (a reduction) $\mathcal{R}$ that, for all $n \in \mathbb{N}$ such that $n = l'(l(n))$ (which are infinity many), distinguishes oracle access to f_k (for uniform $k \xleftarrow{\$} \{0,1\}^{l(n)}$) from oracle access to a uniform $H \xleftarrow{\$} \mathsf{Func}_n$ with nonnegligible advantage. On input 1^n, $\mathcal{R}$ makes queries $0_2, 1_2, \ldots, (p(n)-1)_2$ to the challenger, receives $y_0 = \mathcal{O}(0_2), y_1 = \mathcal{O}(1_2), \ldots, y_{p(n)-1} = \mathcal{O}((p(n)-1)_2)$ as the answers, runs $\mathbf{Adv}^{OW}$ on input $(1^{l(n)}, (y_0, \ldots, y_{p(n)-1}))$, and receives as output $k' \in \{0,1\}^{l(n)}$ from $\mathbf{Adv}^{OW}$. Finally, $\mathcal{R}$ outputs 0 if $g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1})$, and a uniformly random bit otherwise. We illustrate $\mathcal{R}$ in the following distinguishing game G' of oracle accesses to f_{U_n} and U_{Func_n}.

$$
\begin{array}{lccc}
\mathsf{Ch} & & \mathcal{R} & & \mathbf{Adv}^{OW} \\
b \xleftarrow{\$} \{0,1\} & & & & \\
k \xleftarrow{\$} \{0,1\}^{l(n)} & & & & \\
H \xleftarrow{\$} \mathsf{Func}_n & & & & \\
\mathcal{O} := \begin{cases} f_k, & b = 0 \\ H, & b = 1 \end{cases} & & & & \\
& \xrightarrow{1^n} & & & \\
& \xleftarrow{0_2} & & & \\
& \xrightarrow{y_0 = \mathcal{O}(0_2)} & & & \\
& \vdots & & & \\
& \xleftarrow{(p(n)-1)_2} & & & \\
& \xrightarrow{y_{p(n)-1} = \mathcal{O}((p(n)-1)_2)} & & & \\
& & & \xrightarrow{(1^{l(n)}, (y_0, \ldots, y_{p(n)-1}))} & \\
& & & \xleftarrow{k'} & \\
& & b' \begin{cases} := 0, & g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \\ \xleftarrow{\$} \{0,1\}, & g_{l(n)}(k') \neq (y_0, \ldots, y_{p(n)-1}) \end{cases} & & \\
& \xleftarrow{b'} & & & \\
& & \mathcal{R} \text{ wins if } b' = b & &
\end{array}
$$

Figure 5: Game G'

Clearly $\mathcal{R}$ runs in PPT since (i) $\mathbf{Adv}^{OW}$ runs in time $\mathsf{poly}(l(n)) = \mathsf{poly}(\Theta(n^c)) = O(\mathsf{poly}(n))$, (ii) $g_{l(n)}$ can be computed in time $\mathsf{poly}(l(n)) = \mathsf{poly}(\Theta(n^c)) = O(\mathsf{poly}(n))$ (as we have shown that g_m is polynomial-time computable), (iii) f_k can be computed in $\mathsf{poly}(n)$ time (as f is a PRF), and (iv) $\mathcal{R}$ makes $p(n) = O(\mathsf{poly}(n))$ queries. Summing up the run time of each component, we can verify that $\mathcal{R}$ in total runs in time

$$\mathsf{poly}(n) \cdot O(\mathsf{poly}(n)) + O(\mathsf{poly}(n)) + O(\mathsf{poly}(n)) = O(\mathsf{poly}(n)).$$

To see that for all $n \in \mathbb{N}$ such that $n = l'(l(n))$, $\mathcal{R}$ distinguishes oracle accesses to f_{U_n} and U_{Func_n} with nonnegligible advantage, we instead show that $\mathcal{R}$ wins G' with nonnegligible advantage for these n's. Observe that for all $n \in \mathbb{N}$ such that $n = l'(l(n))$:

- If $b = 0$ and thus $\mathcal{O} = f_k$ for uniform $k \xleftarrow{\$} \{0,1\}^{l(n)}$, then

$$
\begin{aligned}
& g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \\
\iff & g_{l(n)}(\mathbf{Adv}^{OW}(1^{l(n)}, (y_0, \ldots, y_{p(n)-1}))) = (y_0, \ldots, y_{p(n)-1}) \\
\iff & g_{l(n)}(\mathbf{Adv}^{OW}(1^{l(n)}, (f_k(0_2), \ldots, f_k((p(n)-1)_2)))) = (f_k(0_2), \ldots, f_k((p(n)-1)_2)) \\
\iff & g_{l(n)}(\mathbf{Adv}^{OW}(1^{l(n)}, g_{l(n)}(k))) = g_{l(n)}(k) \\
\iff & \mathbf{Adv}^{OW} \text{ wins } \mathsf{G} \text{ with } m := l(n).
\end{aligned}
$$

We can therefore calculate the win rate of $\mathcal{R}$ when $b = 0$:

$$
\begin{aligned}
&\Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[\mathcal{R}^{f_k} \text{ wins } \mathsf{G}' \;\middle|\; b = 0 \right] \\
=& \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} [b' = 0 \mid b = 0] \\
=& \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \mid b = 0 \right] \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} [0 = 0 \mid b = 0] \\
&\quad + \left(1 - \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \mid b = 0 \right] \right) \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} [U_1 = 0 \mid b = 0] \\
=& \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[\mathbf{Adv}^{OW} \text{ wins } \mathsf{G} \text{ with } m \coloneqq l(n) \right] \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} [0 = 0 \mid b = 0] \\
&\quad + \left(1 - \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[\mathbf{Adv}^{OW} \text{ wins } \mathsf{G} \text{ with } m \coloneqq l(n) \right] \right) \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} [U_1 = 0 \mid b = 0] \\
=& \varepsilon(l(n)) \cdot 1 + (1 - \varepsilon(l(n))) \cdot \frac{1}{2} = \frac{1}{2} + \frac{\varepsilon(l(n))}{2}.
\end{aligned}
$$

- If $b = 1$ and thus $\mathcal{O} = H$ for uniform $H \xleftarrow{\$} \mathsf{Func}_n$, observe that

$$
\begin{aligned}
& g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \\
\iff & (f_{k'}(0_2), \ldots, f_{k'}((p(n)-1)_2)) = (H(0_2), \ldots, H((p(n)-1)_2)) \\
\implies & \exists k^* \in \{0,1\}^{l(n)}.\ (f_{k^*}(0_2), \ldots, f_{k^*}((p(n)-1)_2)) = (H(0_2), \ldots, H((p(n)-1)_2)) \\
\iff & (H(0_2), \ldots, H((p(n)-1)_2)) \in \left\{ (f_{k^*}(0_2), \ldots, f_{k^*}((p(n)-1)_2)) \right\}_{k^* \in \{0,1\}^{l(n)}} = \left\{ g_{l(n)}(k^*) \right\}_{k^* \in \{0,1\}^{l(n)}} \\
& = \mathrm{im}(g_{l(n)}).
\end{aligned}
$$

Since H is uniformly chosen from Func_n, $\{H(x)\}_{x \in \{0,1\}^n}$ are mutually independent and uniformly random. As a result, $(H(0_2), \ldots, H((p(n)-1)_2))$, regardless of the inputs of H, is uniformly distributed over $\{0,1\}^{n \times p(n)}$. We can therefore bound from above $\Pr[g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \mid b = 1]$ by

$$
\begin{aligned}
\Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \mid b = 1 \right] \leq & \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[(H(0_2), \ldots, H((p(n)-1)_2)) \in \mathrm{im}(g_{l(n)}) \right] \\
=& \frac{\left| \mathrm{im}(g_{l(n)}) \right|}{\left| \{0,1\}^{n \times p(n)} \right|} \\
\leq& \frac{\left| \mathrm{dom}(g_{l(n)}) \right|}{\left| \{0,1\}^{n \times p(n)} \right|} \\
=& \frac{2^{l(n)}}{2^{n \times p(n)}} = 2^{-(np(n) - l(n))}.
\end{aligned}
\tag{3.1}
$$

Finally, we calculate the win rate of $\mathcal{R}$ when $b = 1$:

$$
\begin{aligned}
&\Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[\mathcal{R}^H \text{ wins } \mathsf{G}' \;\middle|\; b = 1\right] \\
=& \Pr_{H \xleftarrow{\$} \mathsf{Func}_n}[b' = 1 \mid b = 1] \\
=& \Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \mid b = 1\right] \cdot 0 \\
&+ \left(1 - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[g_{l(n)}(k') = (y_0, \ldots, y_{p(n)-1}) \mid b = 1\right]\right) \cdot \frac{1}{2} \\
\geq& 0 + \frac{1}{2}(1 - 2^{-(np(n)-l(n))}) \qquad \text{(by (3.1))} \\
\geq& \frac{1}{2} - 2^{-(np(n)-l(n))}.
\end{aligned}
$$

Combining the two cases together, we have:

$$
\begin{aligned}
&\Pr_{b \xleftarrow{\$} \{0,1\}, k \xleftarrow{\$} \{0,1\}^{l(n)}, H \xleftarrow{\$} \mathsf{Func}_n}\left[\mathcal{R}^H \text{ wins } \mathsf{G}'\right] \\
=& \Pr_{b \xleftarrow{\$} \{0,1\}}[b = 0] \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}}\left[\mathcal{R}^{f_k} \text{ wins } \mathsf{G}' \;\middle|\; b = 0\right] + \Pr_{b \xleftarrow{\$} \{0,1\}}[b = 1] \Pr_{H \xleftarrow{\$} \mathsf{Func}_n}\left[\mathcal{R}^H \text{ wins } \mathsf{G}' \;\middle|\; b = 1\right] \\
\geq& \frac{1}{2} \cdot \left(\frac{1}{2} + \frac{\varepsilon(l(n))}{2}\right) + \frac{1}{2} \cdot \left(\frac{1}{2} - 2^{-(np(n)-l(n))}\right) \\
\geq& \frac{1}{2} + \left(\frac{1}{4}\varepsilon(l(n)) - 2^{-(np(n)-l(n))}\right).
\end{aligned}
$$

for all $n \in \mathbb{N}$ such that $n = l'(l(n))$ (of which there are infinitely many). Notice that since $\varepsilon(n)$ is nonnegligible and $l(n) = \Theta(n^c) = O(\mathsf{poly}(n))$, $\varepsilon(l(n))$ is also nonnegligible. On the other hand, if we have $p(n) = \Omega(n^c)$, then $2^{-(np(n)-l(n))}$ is negligible because

$$
2^{-(np(n)-l(n))} = 2^{-(n \cdot \Theta(n^c) - \Theta(n^c))} = 2^{-(\Theta(n^{c+1}) - \Theta(n^c))} = 2^{-\Theta(n^{c+1})} = 2^{-\omega(\log n)} = o(n^{-c^\star})
$$

for all $c^\star > 0$. Thus, we set $p(n) \triangleq n^c$. (Note that we also require in the definition of g_m that $p(n) = O(\mathsf{poly}(n))$.) Finally, since $\frac{1}{4}\varepsilon(l(n))$ is nonnegligible whereas $2^{-(np(n)-l(n))}$ is negligible, the total advantage $\frac{1}{4}\varepsilon(l(n)) - 2^{-(np(n)-l(n))}$ of $\mathcal{R}$ is thus nonnegligible.

Hence, the existence of the PPT oracle distinguisher $\mathcal{R}$ implies f is not a PRF, contradicting the assumption. We have thus completed the proof of (2), finishing the whole proof. ⊛

Answer (b). Assume that there exists a PRG $G_n : \{0,1\}^n \to \{0,1\}^{l(n)}$. By expanding/truncating G_n, we can construct a PRG $G'_n : \{0,1\}^n \to \{0,1\}^{2n}$, which we will work with in this subproblem. We show that G'_n is also a OWF. To show this, there are two properties to prove: (1) G'_n is polynomial-time computable, and (2) G'_n can be inverted with only negligible probability by any PPT adversary. Note that (1) trivially holds as G'_n is also a PRG, so it remains to show (2).

We prove (2) by contradiction. Suppose to the contrary that there exists a PPT adversary $\mathbf{Adv}^{OW}$ that wins the following inverting game of G'_n, denoted G, with probability $\varepsilon(n)$ for all $n \in \mathbb{N}$ where ε

is nonnegligible.

$$\begin{array}{ccc} \textsf{Ch} & & \mathbf{Adv}^{OW} \\ x \leftarrow \{0,1\}^n & & \\ & \xrightarrow{(1^n, y := G'_n(x))} & \\ & \xleftarrow{x'} & \end{array}$$

$\mathbf{Adv}^{OW}$ wins if $G'_n(x') = G'_n(x)$

Figure 6: Game G

By abusing $\mathbf{Adv}^{OW}$, we construct a PPT distinguisher (a reduction) $\mathcal{R}$ that distinguishes between $\{G'_n(U_n)\}_{n\in\mathbb{N}}$ and $\{U_{2n}\}_{n\in\mathbb{N}}$ with nonnegligible advantage for all $n \in \mathbb{N}$, as shown in the following distinguishing game G′ of $\{G'_n(U_n)\}_{n\in\mathbb{N}}$ and $\{U_{2n}\}_{n\in\mathbb{N}}$. Namely, on input $y \in \{0,1\}^{2n}$, $\mathcal{R}$ runs $\mathbf{Adv}^{OW}$ on input $(1^n, y)$ and receives as output x' from $\mathbf{Adv}^{OW}$. Finally, $\mathcal{R}$ outputs 0 if $G'_n(x') = y$, and a uniformly random bit otherwise.

$$\begin{array}{ccccc} \textsf{Ch} & & \mathcal{R} & & \mathbf{Adv}^{OW} \\ b \xleftarrow{\$} \{0,1\} & & & & \\ y \leftarrow \begin{cases} G'_n(U_n), & b = 0 \\ U_{2n}, & b = 1 \end{cases} & & & & \\ & \xrightarrow{y} & & & \\ & & & \xrightarrow{(1^n, y)} & \\ & & & \xleftarrow{x'} & \\ & & b' \begin{cases} := 0, & G'_n(x') = y \\ \xleftarrow{\$} \{0,1\}, & G'_n(x') \neq y \end{cases} & & \\ & \xleftarrow{b'} & & & \end{array}$$

$\mathcal{R}$ wins if $b' = b$

Figure 7: Game G′

Clearly, $\mathcal{R}$ is a PPT algorithm since $\mathbf{Adv}^{OW}$ is a PPT algorithm. We are now left to show that $\mathcal{R}$ wins G′ with nonnegligible advantage. To see this, observe that for all $n \in \mathbb{N}$

$$\Pr_{y \leftarrow G'_n(U_n)} [G'_n(x') = y] = \Pr_{x \xleftarrow{\$} \{0,1\}^n} [G'_n(x') = G'_n(x)] = \Pr_{x \xleftarrow{\$} \{0,1\}^n} \left[\mathbf{Adv}^{OW} \text{ wins G}\right] = \varepsilon(n) \tag{3.2}$$

and that for all $n \in \mathbb{N}$

$$\Pr_{y \leftarrow U_{2n}} [G'_n(x') = y] = \Pr_{y \leftarrow U_{2n}} \left[G'_n(\mathbf{Adv}^{OW}(1^n, y)) = y\right] = \Pr_{y \leftarrow U_{2n}} [y \in \mathrm{im}(G'_n)] = \frac{|\mathrm{im}(G'_n)|}{|\{0,1\}^{2n}|} \leq \frac{|\mathrm{dom}(G'_n)|}{|\{0,1\}^{2n}|} = \frac{2^n}{2^{2n}} = 2^{-n}. \tag{3.3}$$

We then calculate a lower bound of the advantage of $\mathcal{R}$ in G', for all $n \in \mathbb{N}$, as follows:

$$
\begin{aligned}
&\Pr_{b \xleftarrow{\$} \{0,1\}, y \leftarrow \begin{cases} G'_n(U_n), & b=0 \\ U_{2n}, & b=1 \end{cases}} [\mathcal{R} \text{ wins } \mathsf{G}'] \\
=& \Pr_{b \xleftarrow{\$} \{0,1\}, y \leftarrow \begin{cases} G'_n(U_n), & b=0 \\ U_{2n}, & b=1 \end{cases}} [b' = b] \\
=& \Pr_{b \xleftarrow{\$} \{0,1\}} [b=0] \Pr_{y \leftarrow G'_n(U_n)} [b'=0 \mid b=0] + \Pr_{b \xleftarrow{\$} \{0,1\}} [b=1] \Pr_{y \leftarrow U_{2n}} [b'=1 \mid b=1] \\
=& \frac{1}{2} \Pr_{y \leftarrow G'_n(U_n)} [b'=0] + \frac{1}{2} \Pr_{y \leftarrow U_{2n}} [b'=1] \\
=& \frac{1}{2} \left(\Pr_{y \leftarrow G'_n(U_n)} [G'_n(x') = y] \cdot 1 + \left(1 - \Pr_{y \leftarrow G'_n(U_n)} [G'_n(x') = y] \right) \cdot \frac{1}{2} \right) \\
&\quad + \frac{1}{2} \left(\Pr_{y \leftarrow U_{2n}} [G'_n(x') = y] \cdot 0 + \left(1 - \Pr_{y \leftarrow U_{2n}} [G'_n(x') = y] \right) \cdot \frac{1}{2} \right) \\
\geq& \frac{1}{2} \left(\varepsilon(n) \cdot 1 + (1 - \varepsilon(n)) \cdot \frac{1}{2} \right) + \frac{1}{2} \left(0 + \left(1 - 2^{-n}\right) \cdot \frac{1}{2} \right) \qquad \text{(by (3.2) and (3.3))} \\
=& \frac{1}{2} + \left(\frac{1}{4} \varepsilon(n) - 2^{-n-2} \right).
\end{aligned}
$$

Notice that $\frac{1}{4}\varepsilon(n)$ is nonnegligible whereas 2^{-n-2} is negligible, so $\frac{1}{4}\varepsilon(n) - 2^{-n-2}$ is nonnegligible.

As shown above, the PPT distinguisher $\mathcal{R}$ distinguishes between $\{G'_n(U_n)\}_{n \in \mathbb{N}}$ and $\{U_{2n}\}_{n \in \mathbb{N}}$ with nonnegligible advantage for all $n \in \mathbb{N}$. This implies $\{G'_n(U_n)\}_{n \in \mathbb{N}}$ is not pseudorandom and thus G'_n is not a PRG, contradicting the fact that G'_n is. We have thus completed the proof of (2), finishing the whole proof. ⊛

Answer (c). Assume that there exists a PRF $f : \{0,1\}^{l(n)} \times \{0,1\}^n \to \{0,1\}^n$ where $l(n) = \Theta(n^c)$ for some constant $c > 0$. As before, define an "inverse" function l' of l by

$$
l'(m) \triangleq \min\{n \in \mathbb{N} \mid l(n) = m\}.
$$

Note that we have already proven in Claim 3.1 that $l'(m) = \Theta(m^{c'})$ for some $c' > 0$.

We show that the function $G_m : \{0,1\}^m \to \{0,1\}^{m+1}$ defined by

$$
G_m(k) \triangleq (f_k(0_2))_{[1]} \parallel (f_k(1_2))_{[1]} \parallel \ldots \parallel (f_k((m)_2))_{[1]},
$$

where $(x)_2$ denotes the $l'(m)$-bit representation of x and $(x)_{[i]}$ denotes the ith bit of x, is a PRG. Note that we define G_m for only those $m \in \mathbb{N}$ such that m is representable in $l'(m)$ bits (i.e., $m \leq 2^{l'(m)} - 1$), which holds for sufficiently large m since $l'(m) = \Theta(m^{c'})$.

To show this, there are three properties to prove: (1) G_m is deterministic and polynomial-time computable, (2) $|G_m(x)| > |x|$ for all x, and (3) $\{G_m(U_m)\}_{m \in \mathbb{N}}$ is pseudorandom.

Firstly, to see (1) holds, note that since $l'(m) = \Theta(m^{c'})$ and $f_k : \{0,1\}^n \to \{0,1\}^n$ can be computed in $\mathsf{poly}(n)$ time (as f is a PRF) where $n = l'(m)$, G_m can be computed in time $\mathsf{poly}(l'(m))(m+1) = O(\mathsf{poly}(m))$, as desired. Besides, (2) is trivial.

To see (3) holds, we prove it by contradiction. Suppose to the contrary that there exists a PPT distinguisher D such that

$$\varepsilon(m) \triangleq \left|\Pr\left[D(G_m(U_m)) = 1\right] - \Pr\left[D(U_{m+1}) = 1\right]\right| \tag{3.4}$$

is nonnegligible. Equivalently speaking, D wins the following distinguishing game of $\{G_m(U_m)\}_{m\in\mathbb{N}}$ and $\{U_{m+1}\}_{m\in\mathbb{N}}$, denoted G, with probability $\frac{1}{2} + \frac{1}{2}\varepsilon(m)$ for all $m \in \mathbb{N}$, where ε is nonnegligible.

$$\begin{array}{ccc}
\mathsf{Ch} & & D \\
b \leftarrow \{0,1\} & & \\
y \leftarrow \begin{cases} G_m(U_m), & b = 0 \\ U_{m+1}, & b = 1 \end{cases} & & \\
& \xrightarrow{\quad y \quad} & \\
& \xleftarrow{\quad b' \quad} & \\
& D \text{ wins if } b' = b &
\end{array}$$

Figure 8: Game G

By abusing D, we construct a PPT oracle distinguisher (a reduction) $\mathcal{R}$ that, for all $n \in \mathbb{N}$ such that $n = l'(l(n))$ (which are infinity many), distinguishes oracle access to f_k (for uniform $k \xleftarrow{\$} \{0,1\}^{l(n)}$) from oracle access to a uniform $H \xleftarrow{\$} \mathsf{Func}_n$ with nonnegligible advantage, as shown in the following distinguishing game G' of oracle accesses to f_{U_n} and U_{Func_n} Namely, on input 1^n, $\mathcal{R}$ makes queries $0_2, 1_2, \ldots, (l(n))_2$ to the challenger, receives as the answers $y_0 = \mathcal{O}(0_2), y_1 = \mathcal{O}(1_2), \ldots, y_{l(n)} = \mathcal{O}(l(n))_2)$, runs D on input $(y_0)_{[1]} \parallel \ldots \parallel (y_{l(n)})_{[1]}$. When D terminates, $\mathcal{R}$ receives as output a bit b' from D and directly outputs b'.

$$\begin{array}{ccccc}
\mathsf{Ch} & & \mathcal{R} & & D \\
b \xleftarrow{\$} \{0,1\} & & & & \\
k \xleftarrow{\$} \{0,1\}^{l(n)},\ H \xleftarrow{\$} \mathsf{Func}_n & & & & \\
\mathcal{O} := \begin{cases} f_k, & b = 0 \\ H, & b = 1 \end{cases} & & & & \\
& \xrightarrow{\quad 1^n \quad} & & & \\
& \xleftarrow{\quad 0_2 \quad} & & & \\
& \xrightarrow{\ y_0 = \mathcal{O}(0_2)\ } & & & \\
& \vdots & & & \\
& \xleftarrow{\quad (l(n))_2 \quad} & & & \\
& \xrightarrow{\ y_{l(n)} = \mathcal{O}((l(n))_2)\ } & & & \\
& & & \xrightarrow{\ (y_0)_{[1]} \| \cdots \| (y_{l(n)})_{[1]}\ } & \\
& & & \xleftarrow{\quad b' \quad} & \\
& \xleftarrow{\quad b' \quad} & & & \\
& \mathcal{R} \text{ wins if } b' = b & & &
\end{array}$$

Figure 9: Game G'

Clearly $\mathcal{R}$ runs in PPT since (i) D runs in time $\mathsf{poly}(l(n)) = \mathsf{poly}(\Theta(n^c)) = O(\mathsf{poly}(n))$, (ii) f_k can be computed in $\mathsf{poly}(n)$ time (as f is a PRF), and (iii) $l(n) + 1 = \Theta(n^c) + 1 = O(\mathsf{poly}(n))$ queries are made. Summing up the run time of each component, we can verify that $\mathcal{R}$ in total runs in time

$$\mathsf{poly}(n) \cdot O(\mathsf{poly}(n)) + O(\mathsf{poly}(n)) = O(\mathsf{poly}(n)).$$

To see that for all $n \in \mathbb{N}$ such that $n = l'(l(n))$, $\mathcal{R}$ distinguishes oracle accesses to f_{U_n} and U_{Func_n} with nonnegligible advantage, observe that for all $n \in \mathbb{N}$ such that $n = l'(l(n))$,

$$\begin{aligned}
\Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[\mathcal{R}^{f_k}(1^n) = 1\right] &= \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[D\left((f_k(0_2))_{[1]} \parallel \ldots \parallel (f_k((l(n))_2))_{[1]}\right) = 1\right] \\
&= \Pr_{k \xleftarrow{\$} \{0,1\}^{l(n)}} \left[D\left(G_{l(n)}(k)\right) = 1\right] \\
&= \Pr\left[D\left(G_{l(n)}(U_{l(n)})\right) = 1\right], \qquad (3.5)
\end{aligned}$$

and

$$\begin{aligned}
\Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[\mathcal{R}^{H}(1^n) = 1\right] &= \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[D\left((H(0_2))_{[1]} \parallel \ldots \parallel (H((l(n))_2))_{[1]}\right) = 1\right] \\
&= \Pr\left[D\left((U_n^{(1)})_{[1]} \parallel \ldots \parallel (U_n^{(l(n)+1)})_{[1]}\right) = 1\right] \\
&= \Pr\left[D\left(U_1^{(1)} \parallel \ldots \parallel U_1^{(l(n)+1)}\right) = 1\right] \\
&= \Pr\left[D\left(U_{l(n)+1}\right) = 1\right]. \qquad (3.6)
\end{aligned}$$

Hence, we calculate the advantage of $\mathcal{R}$ in distinguishing between oracle accesses to f_{U_n} and U_{Func_n} as follows:

$$\begin{aligned}
&\left|\Pr_{k \xleftarrow{\$} l(n)} \left[\mathcal{R}^{f_k}(1^n) = 1\right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n} \left[\mathcal{R}^{H}(1^n) = 1\right]\right| \\
=&\left|\Pr\left[D\left(G_{l(n)}(U_{l(n)})\right) = 1\right] - \Pr\left[D\left(U_{l(n)+1}\right) = 1\right]\right| && \text{(by (3.5) and (3.6))} \\
=&\varepsilon(l(n)) && \text{(by (3.4))}
\end{aligned}$$

for all $n \in \mathbb{N}$ such that $n = l'(l(n))$ (of which there are infinitely many). Notice that since $\varepsilon(n)$ is nonnegligible and $l(n) = \Theta(n^c) = O(\mathsf{poly}(n))$, the total advantage $\varepsilon(l(n))$ of $\mathcal{R}$ is also nonnegligible.

Hence, the existence of the PPT oracle distinguisher $\mathcal{R}$ implies f is not a PRF, contradicting the assumption. We have thus completed the proof of (3), finishing the whole proof. ⊛

Problem 4 (Extending GGM: Larger Range and Flexible Inputs).

Let us first recall the GGM construction. Given a PRG $G : \{0,1\}^n \to \{0,1\}^{2n}$, where we define $G_0, G_1 : \{0,1\}^n \to \{0,1\}^n$ such that $G(\cdot) = G_0(\cdot) \parallel G_1(\cdot)$, we learned that we can construct an PRF $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ where

$$F_k(x) = (G_{x_n} \circ G_{x_{n-1}} \circ \cdots \circ G_{x_1})(k).$$

In the GGM construction described above, the PRF F_k maps n-bit inputs to n-bit outputs. However, in general, there is no requirement that a PRF must restrict both its input and output lengths to n bits. It is straightforward to extend the above construction to support inputs of length $t(n)$, for some polynomial $t(\cdot)$, as follows:

$$F_k(x) = (G_{x_{t(n)}} \circ G_{x_{t(n)-1}} \circ \cdots \circ G_{x_1})(k). \tag{4.1}$$

The security of the above construction is identical to that of the original case. However, even in this extension, F_k only supports inputs of a fixed length $x \in \{0,1\}^{t(n)}$, and the output length remains n bits. In the following problem, you are asked to construct PRFs that support multiple input lengths, where each input length may correspond to a different output length. Note that the security definition of a PRF with varying input and output lengths can be naturally extended by considering a random function with appropriately defined input and output domains.

(a) The first attempt is to use the same construction to directly support multiple input lengths:

$$F_k(x) = (G_{x_{|x|}} \circ G_{x_{|x|-1}} \circ \cdots \circ G_{x_1})(k).$$

Show that the above construction is not secure, even when it supports only two input lengths, namely $\{0,1\}^n$ and $\{0,1\}^{n-1}$.

(b) We say PRF F_k support (a,b) query if for any $x \in \{0,1\}^a$, then $F_k(x) \in \{0,1\}^b$. Construct a PRF that can support an arbitrary $\ell(n)$ distinct (a_i, b_i) queries, where ℓ is some polynomial and for all $i \in [\ell(n)]$, both a_i and b_i are also bounded by some polynomial in n. (You may assume the existence of a pseudorandom generator (PRG) with arbitrary polynomial stretch and construction in Equation (4.1) is a secure PRF.)

Answer (a). First, we have to adjust the definition of PRF for keyed functions of arbitrary-length inputs (F in particular). Let $\mathsf{Func}_n^* \triangleq (\{0,1\}^n)^{\{0,1\}^*}$, which denotes the set of all functions mapping strings of any length to n-bit strings.

Definition 4.1 (PRF). A keyed function $f : \{0,1\}^{\ell(n)} \times \{0,1\}^* \to \{0,1\}^n$ is a PRF if

- f can be computed in deterministic polynomial time (i.e. in time $O(\mathsf{poly}(n))$), and
- for all PPT oracle distinguishers D, there exists a negligible function ε such that for every $n \in \mathbb{N}$

$$\left| \Pr_{k \xleftarrow{\$} \{0,1\}^{\ell(n)}} \left[D^{f_k}(1^n) = 1 \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n^*} \left[D^H(1^n) = 1 \right] \right| \leq \varepsilon(n).$$

We prove $F : \{0,1\}^n \times \{0,1\}^* \to \{0,1\}^n$ is not a PRF by giving a PPT oracle distinguisher D that, for all $n \in \mathbb{N}$, distinguishes oracle access to F_k (for uniform $k \xleftarrow{\$} \{0,1\}^n$) from that to a uniformly chosen function $H \xleftarrow{\$} \mathsf{Func}_n^*$ with nonnegligible advantage, i.e., wins the PRF game $\mathsf{G}_{\mathsf{PRF}}$ of F with nonnegligible advantage. We construct D as follows:

On input 1^n, make queries $0^{n-1}, 0^n$ to the challenger, receiving as answers $\mathcal{O}(0^{n-1}), \mathcal{O}(0^n)$. Then, check whether $G_0(\mathcal{O}(0^{n-1})) = \mathcal{O}(0^n)$. Output 0 if true, and 1 otherwise.

We illustrate D in the PRF game $\mathsf{G}_{\mathsf{PRF}}$ of F as shown below.

$$
\begin{array}{ccc}
\textsf{Ch} & & D \\
b \xleftarrow{\$} \{0,1\} & & \\
k \xleftarrow{\$} \{0,1\}^n,\ H \xleftarrow{\$} \textsf{Func}^*_n & & \\
\mathcal{O} := \begin{cases} F_k, & b = 0 \\ H, & b = 1 \end{cases} & \xrightarrow{\quad 1^n \quad} & \\
& \xleftarrow{\quad 0^{n-1} \quad} & \\
& \xrightarrow{\quad \mathcal{O}(0^{n-1}) \quad} & \\
& \xleftarrow{\quad 0^{n} \quad} & \\
& \xrightarrow{\quad \mathcal{O}(0^{n}) \quad} & \\
& & b' := \begin{cases} 0 & \text{if } G_0(\mathcal{O}(0^{n-1})) = \mathcal{O}(0^n) \\ 1 & \text{if } G_0(\mathcal{O}(0^{n-1})) \neq \mathcal{O}(0^n) \end{cases} \\
& \xleftarrow{\quad b' \quad} & \\
& D \text{ wins if } b' = b &
\end{array}
$$

Figure 10: Game $\mathsf{G}_{\mathsf{PRF}}$

Clearly D runs in time $O(\mathsf{poly}(n))$ and is thus a PPT algorithm. Observe that in $\mathsf{G}_{\mathsf{PRF}}$, for $n \in \mathbb{N}$:

- If $b = 0$ and $\mathcal{O} := f_k$ for uniform $k \xleftarrow{\$} \{0,1\}^n$, then

$$D \text{ wins} \iff G_0(F_k(0^{n-1})) = F_k(0^n) \iff G_0((\overbrace{G_0 \circ \cdots \circ G_0}^{n-1})(k)) = (\overbrace{G_0 \circ \cdots \circ G_0}^{n})(k) \iff \top,$$

  implying $\Pr_{k \xleftarrow{\$} \{0,1\}^n}[D \text{ wins}] = 1$.

- If $b = 1$ and $\mathcal{O} := H$ for uniform $H \xleftarrow{\$} \textsf{Func}^*_n$, then since $H(0^n) \in \{0,1\}^n$ is uniformly random and independent of $H(0^{n-1}) \in \{0,1\}^n$ and therefore $G_0(H(0^{n-1})) \in \{0,1\}^n$, we have

$$\Pr_{H \xleftarrow{\$} \textsf{Func}^*_n}[D \text{ wins}] = 1 - \Pr_{H \xleftarrow{\$} \textsf{Func}^*_n}\left[G_0(\mathcal{O}(0^{n-1})) = \mathcal{O}(0^n)\right] = 1 - \frac{1}{2^n}.$$

Combining the two together, for $n \in \mathbb{N}$, the win rate of D in $\mathsf{G}_{\mathsf{PRF}}$ is hence

$$
\begin{aligned}
&\Pr_{b \xleftarrow{\$} \{0,1\}, k \xleftarrow{\$} \{0,1\}^n, H \xleftarrow{\$} \textsf{Func}^*_n}[D \text{ wins } \mathsf{G}_{\mathsf{PRF}}] \\
=& \Pr_{b \xleftarrow{\$} \{0,1\}}[b = 0] \Pr_{k \xleftarrow{\$} \{0,1\}^n}[D \text{ wins } \mathsf{G}_{\mathsf{PRF}}] + \Pr_{b \xleftarrow{\$} \{0,1\}}[b = 1] \Pr_{H \xleftarrow{\$} \textsf{Func}^*_n}[D \text{ wins } \mathsf{G}_{\mathsf{PRF}}] \\
=& \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \left(1 - \frac{1}{2^n}\right) = 1 - \frac{1}{2^{n+1}},
\end{aligned}
$$

where the advantage $\frac{1}{2} - \frac{1}{2^{n+1}}$ of D is nonnegligible. Equivalently,

$$\left| \Pr_{k \xleftarrow{\$} \{0,1\}^n} \left[D^{F_k}(1^n) = 1 \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}_n^*} \left[D^H(1^n) = 1 \right] \right| = 1 - \frac{1}{2^n}$$

is nonnegligible. This breaks the pseudorandomness of F defined by Definition 4.1, so F is not a PRF.

⊛

Answer (b). Let $\{(a_i(n), b_i(n))\}_{i\in[\ell(n)]}$ be $\ell(n)$ pairs of polynomially bounded functions for some polynomial ℓ. Assume that $a_i(n) \geq n$ for every $i \in [\ell(n)]$ and $n \in \mathbb{N}$.

Assume that there exists a PRG $G_{m(n)} : \{0,1\}^n \to \{0,1\}^{m(n)}$ of expansion factor $m(n)$ for every polynomially bounded function $m(n) = O(\mathsf{poly}(n))$, and that, given $G_{n\mapsto 2n}$, the GGM construction $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ is indeed a PRF. For $n \in \mathbb{N}$ and $j \in [\ell(n)]$, define

$$\mathsf{Func}^{\bigstar}_{n,j} \triangleq \{f : \{0,1\}^n \to \{0,1\}^{b_j(n)}\}, \quad \mathsf{Func}_n \triangleq \{f : \{0,1\}^n \to \{0,1\}^n\}.$$

Note that by definition, we must have $a_i(n) \neq a_j(n)$ for any $i \neq j$, as otherwise we would have $b_i(n) \neq b_j(n)$ for some $a_i(n) = a_j(n)$ with $i \neq j$, which no PRF could support. Hence for every $n \in \mathbb{N}$, $a_{(\cdot)}(n)$ is injective, meaning that there exists a mapping $a_i(n) \mapsto b_i(n)$ for every $i \in [\ell(n)]$.

We now define a keyed function $f : \{0,1\}^{\sum_{i\in[\ell(n)]} a_i(n)} \times \bigcup_{i\in[\ell(n)]}\{0,1\}^{a_i(n)} \to \bigcup_{i\in[\ell(n)]}\{0,1\}^{b_i(n)}$ for each $n \in \mathbb{N}$ by

$$f_{k_1\|\cdots\|k_{\ell(n)}}(x) \triangleq G_{b_i(n)}\left(F_{k_i}(x)_{[1:n]}\right) \quad \text{for every } x, k_i \in \{0,1\}^{a_i(n)} \text{ and } i \in [\ell(n)].$$

Note that for each $n \in \mathbb{N}$, since $a_{(\cdot)}(n)$ is injective, given an input $x \in \{0,1\}^{a_i(n)}$ for $i \in [\ell(n)]$ to f_k, the value of i is immediately clear from $|x| = a_i(n)$, so f is well-defined.

We claim that (1) f is a PRF that (2) supports $\{(a_i(n), b_i(n))\}_{i\in[\ell(n)]}$ queries. First, assuming f is indeed a PRF, it's easy to see (2) holds since for every $x \in \{0,1\}^{a_i(n)}$ and $k \in \{0,1\}^{\sum_{i\in[\ell(n)]} a_i(n)}$, by the definition of $G_{b_i(n)}$, we have $f_k(x) \in \{0,1\}^{b_i(n)}$. Thus, it remains to show (1) holds.

Suppose to the contrary that f is not a PRF. As f is clearly deterministic and polynomial-time computable, it must be that there exists a PPT oracle distinguisher D and $c > 0$ such that for infinitely many $n \in \mathbb{N}$

$$\left| \Pr_{k_1 \xleftarrow{\$} \{0,1\}^{a_1(n)}, \ldots, k_{\ell(n)} \xleftarrow{\$} \{0,1\}^{a_{\ell(n)}(n)}} \left[D^{f_{k_1\|\cdots\|k_{\ell(n)}}}(1^n) \right] - \Pr_{H \xleftarrow{\$} \mathsf{Func}^{\dagger}_{n,\ell(n)}} \left[D^H(1^n) \right] \right| \geq \frac{1}{n^c}, \tag{4.2}$$

where $\mathsf{Func}^{\dagger}_n \triangleq \left\{ f : \bigcup_{j\in[\ell(n)]}\{0,1\}^{a_j(n)} \to \bigcup_{j\in[\ell(n)]}\{0,1\}^{b_j(n)} \;\middle|\; f(x) \in \{0,1\}^{b_j(n)} \;\; \forall x \in \{0,1\}^{a_j(n)}, j \in [\ell(n)] \right\}$. Define "hybrid distributions" of (deterministic) functions Hyb_i and Hyb'_i for $i = 0, 1, \ldots, \ell(n)$ by

$$\mathsf{Hyb}_i(x) \triangleq \begin{cases} U^{(j)}_{\mathsf{Func}^{\bigstar}_{n,j}}(x_{[1:n]}) & \text{for } x \in \{0,1\}^{a_j(n)} \text{ and } j \in \{1, \ldots, i\} \\ G_{b_j(n)}(x_{[1:n]}) & \text{for } x \in \{0,1\}^{a_j(n)} \text{ and } j \in \{i+1, \ldots, \ell(n)\} \end{cases}$$

$$\mathsf{Hyb}'_i(x) \triangleq \begin{cases} G_{b_j(n)}(U^{(j)}_{\mathsf{Func}_{a_j(n)}}(x)_{[1:n]}) & \text{for } x \in \{0,1\}^{a_j(n)} \text{ and } j \in \{1, \ldots, i\} \\ G_{b_j(n)}(F_{U^{(j)}_{a_j(n)}}(x)_{[1:n]}) & \text{for } x \in \{0,1\}^{a_j(n)} \text{ and } j \in \{i+1, \ldots, \ell(n)\} \end{cases}$$

for every $x \in \bigcup_{i\in[\ell(n)]} \{0,1\}^{a_i(n)}$.

Notice the equivalences of distributions:

$$f_{U_{a_1(n)}\|\cdots\|U_{a_{\ell(n)}(n)}} \equiv \mathsf{Hyb}'_0 \quad \wedge \quad \mathsf{Hyb}'_{\ell(n)} \equiv \mathsf{Hyb}_0 \quad \wedge \quad U_{\mathsf{Func}^{\dagger}_{n,\ell(n)}} \equiv \mathsf{Hyb}_{\ell(n)}.$$

Therefore, by hybrid lemma, either of the following cases holds:

(I) For all $n \in \mathbb{N}$ such that (4.2) holds, there exists $i_n^* \in [\ell(n)]$ such that

$$\left|\Pr\left[D^{\mathsf{Hyb}_{i_n^*}}(1^n)\right] - \Pr\left[D^{\mathsf{Hyb}_{i_n^*-1}}(1^n)\right]\right| \geq \frac{1}{2\ell(n)n^c}.$$

(II) For all $n \in \mathbb{N}$ such that (4.2) holds, there exists $i_n^* \in [\ell(n)]$ such that

$$\left|\Pr\left[D^{\mathsf{Hyb}'_{i_n^*}}(1^n)\right] - \Pr\left[D^{\mathsf{Hyb}'_{i_n^*-1}}(1^n)\right]\right| \geq \frac{1}{2\ell(n)n^c}.$$

If (I) holds, we can construct a PPT reduction $\mathcal{R}_{i_n^*}$ that breaks the pseudorandomness of $(G_{b_{i_n^*}(n)}(U_n^{(1)}), \ldots, G_{b_{i_n^*}(n)}(U_n^{(t(n))}))$, where $t(n)$ is a polynomial upper bound on the number of queries D makes to its oracle on input 1^n, as shown in the following game $\mathsf{G}_{\mathrm{IND}}^{i_n^*}$. (Note that although we don't know the value of i_n^*, we can still construct a reduction $\mathcal{R}_i$ for each $i \in [\ell(n)]$ and argue that at least one reduction breaks the pseudorandomness of $(G_{b_{i_n^*}(n)}(U_n^{(1)}), \ldots, G_{b_{i_n^*}(n)}(U_n^{(t(n))}))$.)

Ch $\qquad\qquad$ $\mathcal{R}_i$ $\qquad\qquad$ D

$b \xleftarrow{\$} \{0,1\}$

$$y_1, \ldots, y_{t(n)} \leftarrow \begin{cases} G_{b_i(n)}(U_n) & \text{if } b = 0 \\ U_{b_i(n)} & \text{if } b = 1 \end{cases} \xrightarrow{(y_1,\ldots,y_{t(n)})} \qquad t' := 1$$

$$H_0 \xleftarrow{\$} \mathsf{Func}^{\bigstar}_{n,0}, \ldots, H_{i-1} \xleftarrow{\$} \mathsf{Func}^{\bigstar}_{n,i-1} \xrightarrow{1^n}$$

$$\vdots$$

$$\xleftarrow{x_q \in \{0,1\}^{a_{j_q}(n)}}$$

$$y'_q := \begin{cases} H_{j_q}(x_{[1:n]}) & \text{if } j_q < i \\ y_{t'} & \text{if } j_q = i \\ G_{b_{j_q}(n)}(x_{[1:n]}) & \text{if } j_q > i \end{cases}$$

$$t' := t' + 1$$

$$\xrightarrow{y'_q}$$

$$\vdots$$

$$\xleftarrow{b'} \qquad\qquad \xleftarrow{b'}$$

$\mathcal{R}_i$ wins if $b' = b$

Figure 11: Game $\mathsf{G}_{\mathrm{IND}}^{i}$

Observe that for all $n \in \mathbb{N}$ such that (4.2) holds,

$$\begin{aligned} &\left|\Pr\left[\mathcal{R}_i(G_{b_i(n)}(U_n^{(1)}), \ldots, G_{b_i(n)}(U_n^{(t(n))})) = 1\right] - \Pr\left[\mathcal{R}_i(U_{b_i(n)}^{(1)}, \ldots, U_{b_i(n)}^{(t(n))}) = 1\right]\right| \\ =&\left|\Pr\left[D^{\mathsf{Hyb}_{i-1}}(1^n)\right] - \Pr\left[D^{\mathsf{Hyb}_i}(1^n) = 1\right]\right| \\ \overset{\text{if } i=i_n^*}{\geq}& \frac{1}{2\ell(n)n^c} \qquad \text{(by (I))}, \end{aligned}$$

which is nonnegligible as $\ell(n) = \mathsf{poly}(n)$. This means that for all $n \in \mathbb{N}$ such that (4.2) holds (of which there are infinitely many), there always exists some $i_n^* \in [\ell(n)]$ such that $\mathcal{R}_{i_n^*}$ breaks the pseudorandomness of $(G_{b_{i_n^*}(n)}(U_n^{(1)}), \dots, G_{b_{i_n^*}(n)}(U_n^{(t(n))}))$. (Note that $b_{i_n^*}(n)$ is simply a function of n.) Thus, by definition, $(G_{b_{i_n^*}(n)}(U_n^{(1)}), \dots, G_{b_{i_n^*}(n)}(U_n^{(t(n))})) \not\approx_c (U_{b_{i_n^*}(n)}^{(1)}, \dots, U_{b_{i_n^*}(n)}^{(t(n))})$. However, since $G_{b_{i_n^*}(n)}$ is a PRG, $G_{b_{i_n^*}(n)}(U_n) \approx_c U_{b_{i_n^*}(n)}$, implying that (as proven in class) $(G_{b_{i_n^*}(n)}(U_n^{(1)}), \dots, G_{b_{i_n^*}(n)}(U_n^{(t(n))})) \approx_c (U_{b_{i_n^*}(n)}^{(1)}, \dots, U_{b_{i_n^*}(n)}^{(t(n))})$ as $t(n) = \mathsf{poly}(n)$. This is a contradiction.

> Note. Caveat: Our reasoning is quite sloppy here. Actually, instead of constructing multiple reductions $\mathcal{R}_i$, we have to define a single reduction that uniformly chooses $i \xleftarrow{\$} [\ell(n)]$ to begin with. Otherwise, for each $n \in \mathbb{N}$, the reduction $\mathcal{R}_{i_n^*}$ that breaks pseudorandomness is not guaranteed not be fixed (since i_n^* may be different for different $n \in \mathbb{N}$), which is NOT aligned with the definition of indistinguishability.

On the other hand, if (II) holds, we can still construct a PPT reduction $\mathcal{R}'_{i_n^*}$ that breaks the pseudorandomness of F, as shown in the following PRF game $\mathsf{G}_{\mathrm{PRF}}^{i_n^*}$ of F. (Note again that although we don't know the value of i_n^*, we can still construct a reduction $\mathcal{R}'_i$ for each $i \in [\ell(n)]$ and argue that at least one reduction breaks the pseudorandomness of F.)

Ch		$\mathcal{R}'_i$		D
$b \xleftarrow{\$} \{0,1\}$				
$\mathcal{O} \leftarrow \begin{cases} F_{U_{a_i(n)}} & \text{if } b = 0 \\ U_{\mathsf{Func}_{a_i(n)}} & \text{if } b = 1 \end{cases}$	$\xrightarrow{(\mathcal{O}, 1^{a_i(n)})}$	$H_0 \xleftarrow{\$} \mathsf{Func}_{a_1(n)}, \dots, H_{i-1} \xleftarrow{\$} \mathsf{Func}_{a_{i-1}(n)}$		
		$k_{i+1} \xleftarrow{\$} \{0,1\}^{a_{i+1}(n)}, \dots, k_{\ell(n)} \xleftarrow{\$} \{0,1\}^{a_{\ell(n)}(n)}$	$\xrightarrow{1^n}$	
			$\vdots$	
			$\xleftarrow{x_q \in \{0,1\}^{a_{j_q}(n)}}$	
		$y_q \coloneqq \begin{cases} G_{b_{j_q}(n)}(H_{j_q}(x)_{[1:n]}) & \text{if } j_q < i \\ G_{b_{j_q}(n)}(\mathcal{O}(x)_{[1:n]}) & \text{if } j_q = i \\ G_{b_{j_q}(n)}(F_{k_{j_q}}(x)_{[1:n]}) & \text{if } j_q > i \end{cases}$		
			$\xrightarrow{y_q}$	
			$\vdots$	
	$\xleftarrow{b'}$		$\xleftarrow{b'}$	
		$\mathcal{R}'_i$ wins if $b' = b$		

Figure 12: Game $\mathsf{G}_{\mathrm{PRF}}^{i}$

Observe that for all $n \in \mathbb{N}$ such that (4.2) holds,

$$\begin{aligned} & \left| \Pr\left[{\mathcal{R}'_i}^{F_{U_{a_i(n)}}}(1^{a_i(n)}) = 1 \right] - \Pr\left[{\mathcal{R}'_i}^{U_{\mathsf{Func}_{a_i(n)}}}(1^n) = 1 \right] \right| \\ = & \left| \Pr\left[D^{\mathsf{Hyb}'_{i-1}}(1^n) \right] - \Pr\left[D^{\mathsf{Hyb}'_i}(1^n) = 1 \right] \right| \\ \overset{\text{if } i = i_n^*}{\geq} & \frac{1}{2\ell(n) n^c} \qquad \text{(by (II))}, \end{aligned}$$

which is nonnegligible as $\ell(n) = \mathsf{poly}(n)$. This means that for all $n \in \mathbb{N}$ such that (4.2) holds (of which there are infinitely many), there always exists some $i_n^* \in [\ell(n)]$ such that $\mathcal{R}'_{i_n^*}$ breaks the pseudorandomness of $F : \{0,1\}^{a_{i_n^*}(n)} \times \{0,1\}^{a_{i_n^*}(n)} \to \{0,1\}^{a_{i_n^*}(n)}$. Assume that there are infinitely many $m \in \mathbb{N}$ such that $m = a_{i_n^*}(n)$ for some $n \in \mathbb{N}$ such that (4.2) holds. Thus, by definition, F is not a PRF, which is a contradiction.

Therefore, in either case, there would be a contradiction. So f is indeed a PRF. ⊛

Problem 5 (Complete the proof in the class).

Let $G : \{0,1\}^n \to \{0,1\}^{2n}$ be a PRG. Recall that in order to show that

$$(G(U_n^{(1)}), \ldots, G(U_n^{(t)})) \approx_c (U_{2n}^{(1)}, \ldots, U_{2n}^{(t)})$$

for $t = \mathsf{poly}(n)$, where $\{U_n^{(1)}, \ldots, U_n^{(t)}\}$ (resp. $\{U_{2n}^{(1)}, \ldots, U_{2n}^{(t)}\}$) is an independent collection of copies of U_n (resp. U_{2n}). We first constructed the distribution

$$\mathsf{Hyb}_i \triangleq (U_{2n}^{(1)}, \ldots, U_{2n}^{(i)}, G(U_n^{(i+1)}), G(U_n^{(t)}))$$

for each $i \in [t]$ as hybrids. Suppose to the contrary that there exists a PPT distinguisher $\mathcal{D}$ such that for infinitely many $n \in \mathbb{N}$

$$\left| \Pr\left[\mathcal{D}(G(U_n^{(1)}), \ldots, G(U_n^{(t)})) = 1\right] - \Pr\left[\mathcal{D}(U_{2n}^{(1)}, \ldots, U_{2n}^{(t)}) = 1\right] \right| > \frac{1}{n^c}$$

for some $c > 0$. By Hybrid Lemma there exists an $i \in [t]$ such that for infinitely many $n \in \mathbb{N}$

$$\left| \Pr\left[\mathcal{D}(\mathsf{Hyb}_{i-1}) = 1\right] - \Pr\left[\mathcal{D}(\mathsf{Hyb}_i) = 1\right] \right| > \frac{1}{tn^c}.$$

In the class we then constructed a reduction R, supposing that R knows such an i, to break the pseudo-randomness of G. In fact, there is no reason to assume that R knows i. However one can modify the reduction R' as follows:

(i) When receiving a challenge x from the challenger Ch in the PRG game, sample $i \xleftarrow{\$} [t]$.

(i) Sample $k_1, \ldots, k_{i-1} \leftarrow U_{2n}$, $k_{i+1}, \ldots, k_t \leftarrow U_n$.

(i) Send $(k_1, \ldots, k_{i-1}, x, G(k_{i+1}), \ldots, G(k_t))$ to the distinguisher $\mathcal{D}$.

(i) When receiving the answer b' from $\mathcal{D}$, output b' back to Ch.

Show that such a reduction R' can also break the pseudo-randomness of G with the same advantage as R, i.e.,

$$\Pr[R' \text{ wins}] > \frac{1}{2} + \frac{1}{2tn^c}.$$

Answer. WLOG, suppose $\mathcal{D}$ is such that for infinitely many $n \in \mathbb{N}$,

$$\Pr\left[\mathcal{D}(G(U_n^{(1)}), \ldots, G(U_n^{(t)})) = 1\right] - \Pr\left[\mathcal{D}(U_{2n}^{(1)}, \ldots, U_{2n}^{(t)}) = 1\right] \leq 0. \tag{5.1}$$

Otherwise, we can replace $\mathcal{D}$ with $\overline{\mathcal{D}}$, where $\overline{\mathcal{D}}$ outputs the exact opposite bit as $\mathcal{D}$ (i.e., $\overline{\mathcal{D}}(x) \triangleq \overline{\mathcal{D}(x)}$ for all $x \in \{0,1\}^{2n \times t}$) for those $n \in \mathbb{N}$ such that (5.1) does not hold, and thus the new $\mathcal{D}$ would satisfy (5.1).

It's now easy to see that for infinitely many $n \in \mathbb{N}$, the advantage of R' in winning the PRG game of G is bounded below by

$$
\begin{aligned}
\Pr[R' \text{ wins}] &= \Pr\left[b \xleftarrow{\$} \{0,1\}, x \leftarrow \begin{Bmatrix} G(U_n), & b=0 \\ U_{2n}, & b=1 \end{Bmatrix} : R'(x) = b\right] \\
&= \Pr_{b \xleftarrow{\$} \{0,1\}}[b=0] \Pr_{x \leftarrow G(U_n)}[R'(x)=0] + \Pr_{b \xleftarrow{\$} \{0,1\}}[b=1] \Pr_{x \leftarrow U_{2n}}[R'(x)=1] \\
&= \frac{1}{2} + \frac{1}{2}\left(\Pr_{x \leftarrow U_{2n}}[R'(x)=1] - \Pr_{x \leftarrow G(U_n)}[R'(x)=1]\right) \\
&= \frac{1}{2} + \frac{1}{2}\left(\Pr_{\substack{x \leftarrow U_{2n} \\ i \xleftarrow{\$} [t] \\ k_1,\dots,k_{i-1} \leftarrow U_{2n} \\ k_{i+1},\dots,k_t \leftarrow U_n}}[\mathcal{D}(k_1,\dots,k_{i-1},x,G(k_{i+1}),\dots,G(k_t)) = 1]\right. \\
&\qquad\qquad \left. - \Pr_{\substack{x \leftarrow G(U_n) \\ i \xleftarrow{\$} [t] \\ k_1,\dots,k_{i-1} \leftarrow U_{2n} \\ k_{i+1},\dots,k_t \leftarrow U_n}}[\mathcal{D}(k_1,\dots,k_{i-1},x,G(k_{i+1}),\dots,G(k_t)) = 1]\right) \\
&= \frac{1}{2} + \frac{1}{2}\left(\Pr_{i \xleftarrow{\$} [t]}[\mathcal{D}(\mathsf{Hyb}_i) = 1] - \Pr_{i \xleftarrow{\$} [t]}\left[\mathcal{D}(\mathsf{Hyb}_{i-1}) = 1\right]\right) \\
&= \frac{1}{2} + \frac{1}{2}\left(\sum_{i \in [t]} \frac{1}{t} \Pr[\mathcal{D}(\mathsf{Hyb}_i) = 1] - \sum_{i \in [t]} \frac{1}{t} \Pr\left[\mathcal{D}(\mathsf{Hyb}_{i-1}) = 1\right]\right) \\
&= \frac{1}{2} + \frac{1}{2t}\left(\Pr[\mathcal{D}(\mathsf{Hyb}_t) = 1] - \Pr[\mathcal{D}(\mathsf{Hyb}_0) = 1]\right) \qquad \text{(by telescoping sum (kind of))} \\
&= \frac{1}{2} + \frac{1}{2t}\left(\Pr\left[\mathcal{D}(U_{2n}^{(1)},\dots,U_{2n}^{(t)}) = 1\right] - \Pr\left[\mathcal{D}(G(U_n^{(1)}),\dots,G(U_n^{(t)})) = 1\right]\right) \\
&= \frac{1}{2} + \frac{1}{2t}\left|\Pr\left[\mathcal{D}(G(U_n^{(1)}),\dots,G(U_n^{(t)})) = 1\right] - \Pr\left[\mathcal{D}(U_{2n}^{(1)},\dots,U_{2n}^{(t)}) = 1\right]\right| \qquad \text{(by (5.1))} \\
&> \frac{1}{2} + \frac{1}{2t} \cdot \frac{1}{n^c} \qquad \text{(by assumption)} \\
&= \frac{1}{2} + \frac{1}{2tn^c},
\end{aligned}
$$

as desired. ⊛

Problem 6 (Commitment Scheme from PRG).

Let $G : \{0,1\}^n \to \{0,1\}^{3n}$ be a PRG. The construction of interactive commitment scheme $(\langle S, R\rangle, \mathsf{Open})$ is defined as follows:

Construction of $\langle S(b, 1^n), R(1^n)\rangle \to (d_S, \tau)$:

1. R sample a random string $r \in \{0,1\}^{3n}$ and send r to S.
2. S receives the message r. Then S samples a random string $s \in \{0,1\}^n$ and compute $c \leftarrow \mathsf{Com}(s, r, b)$ by
$$c = \mathsf{Com}(s, r, b) := \begin{cases} G(s) & \text{if } b = 0 \\ G(s) \oplus r & \text{if } b = 1 \end{cases}.$$
Lastly, S sends c back to R and output s as its opening information d_S.
3. The transcript $\tau := (r, c)$ which include the first message r and the second message c.

Construction of $\mathsf{Open}(b, d_S, \tau) \to \{\top, \bot\}$:

1. Parse $\tau := (r, c)$.
2. If $\mathsf{Com}(d_s, r, b) = c$, then output $\top$ and otherwise output $\bot$.

Below are the problem statements.

(a) Show that the above scheme satisfies the correctness.

(b) Show that the above scheme is computational hiding.

(c) Show that the above scheme is statistically binding.

(d) Show that there is no interactive commitment scheme that satisfies both statistical hiding and statistical binding.

Answer (a). We prove this by simply expanding the construction of S, R, and Open. Let $b \in \{0,1\}$ and $n \in \mathbb{N}$. Given $(d_S, \tau) \leftarrow \langle S(b, 1^n), R(1^n)\rangle$, we have

$$d_S = s, \quad \tau = \left(r, c \triangleq \begin{cases} G(s) & \text{if } b = 0 \\ G(s) \oplus r & \text{if } b = 1 \end{cases} \right)$$

for some $s \xleftarrow{\$} \{0,1\}^n$ and $r \xleftarrow{\$} \{0,1\}^{3n}$. Therefore, in $\mathsf{Open}(b, d_S, \tau)$, it always holds that

$$\mathsf{Com}(d_S, r', b) = \mathsf{Com}(s, r, b) \overset{\text{def}}{=} \left\{ \begin{matrix} G(s) & \text{if } b = 0 \\ G(s) \oplus r & \text{if } b = 1 \end{matrix} \right\} = c = c',$$

where $(r', c') := \tau$, in which case Open will output $\top$. This implies that $\mathsf{Open}(b, d_S, \tau)$ always outputs $\top$, i.e., that
$$\Pr\left[\mathsf{Open}(b, d_S, \tau) = \top : (d_S, \tau) \leftarrow \langle S(b, 1^n), R(1^n)\rangle\right] = 1,$$
which, by definition, means that the interactive commitment scheme $(\langle S, R\rangle, \mathsf{Open})$ is correct. ⊛

Answer (b). We prove this by contradiction. Suppose to the contrary that the interactive commitment scheme $(\langle S, R\rangle, \mathsf{Open})$ is not computational hiding. Then, by definition, we know there exist a PPT receiver R^*, internal randomness $r_{R^*} \in \{0,1\}^{3n}$ chosen by R^*, and a PPT distinguisher D such that

$$\begin{aligned}
\varepsilon(n) &\triangleq \left| \Pr_{r_S \xleftarrow{\$} \{0,1\}^n} \left[D\left(\mathsf{view}_{R^*}^{\langle S, R^* \rangle}((0,1^n), 1^n; r_S, r_{R^*}) \right) = 1 \right] - \Pr_{r_S \xleftarrow{\$} \{0,1\}^n} \left[D\left(\mathsf{view}_{R^*}^{\langle S, R^* \rangle}((1,1^n), 1^n; r_S, r_{R^*}) \right) = 1 \right] \right| \\
&= \left| \Pr_{s \xleftarrow{\$} \{0,1\}^n} \left[D\left(1^n, r_{R^*}, (r_{R^*}, G(s))\right) = 1 \right] - \Pr_{s \xleftarrow{\$} \{0,1\}^n} \left[D\left(1^n, r_{R^*}, (r_{R^*}, G(s) \oplus r_{R^*})\right) = 1 \right] \right|
\end{aligned} \tag{6.1}$$

is nonnegligible. Having (6.1), we slightly transform D to get another PPT distinguisher D' such that

$$\varepsilon(n) = \left| \Pr_{s \xleftarrow{\$} \{0,1\}^n} \left[D'\left(r_{R^*}, G(s)\right) = 1 \right] - \Pr_{s \xleftarrow{\$} \{0,1\}^n} \left[D'\left(r_{R^*}, G(s) \oplus r_{R^*}\right) = 1 \right] \right|. \tag{6.2}$$

Notice that we've discarded 1^n since D' can always infer n from either of the two inputs, which is of length $3n$. Equivalently, this means that D' wins the following distinguishing game $\mathsf{G}_{\text{hiding}}$ with probability $\frac{1}{2} + \frac{1}{2}\varepsilon(n)$.

$$\begin{array}{ccc}
\mathsf{Ch} & & D' \\
b \xleftarrow{\$} \{0,1\} & & \\
s \xleftarrow{\$} \{0,1\}^n & & \\
c := \begin{cases} G(s) & \text{if } b = 0 \\ G(s) \oplus r_{R^*} & \text{if } b = 1 \end{cases} & & \\
& \xrightarrow{\quad (r_{R^*}, c) \quad} & \\
& \xleftarrow{\quad b' \quad} & \\
D' \text{ wins if } b' = b & &
\end{array}$$

Figure 13: Game $\mathsf{G}_{\text{hiding}}$

By abusing D', we construct a PPT distinguisher (a reduction) $\mathcal{R}$ that wins the PRG game G_{PRG} of $G : \{0,1\}^n \to \{0,1\}^{3n}$, i.e., that distinguishes between the ensembles $\{G(U_n)\}_{n\in\mathbb{N}}$ and $\{U_{3n}\}_{n\in\mathbb{N}}$), with nonnegligible advantage, as shown below. Namely, on input $y \in \{0,1\}^{3n}$, $\mathcal{R}$ plays as the challenger in $\mathsf{G}_{\text{hiding}}$, but with $G(s)$ (for $s \xleftarrow{\$} \{0,1\}^n$) replaced by y, to interact with D'. When D' terminates, $\mathcal{R}$ receives as output b' from D' and then outputs 0 if $b' = b$ (i.e. D' wins $\mathsf{G}_{\text{hiding}}$), and outputs 1 otherwise. Note that if $d = 0$, then $y = G(x)$ for some $x \xleftarrow{\$} \{0,1\}^n$; if $d = 1$, then $y \leftarrow U_{3n}$

is uniform and independent of r_{R^*}.

$$
\begin{array}{ccccc}
\mathsf{Ch} & & \mathcal{R} & & D' \\
d \xleftarrow{\$} \{0,1\} & & & & \\
y \leftarrow \begin{cases} G(U_n) & \text{if } d = 0 \\ U_{3n} & \text{if } d = 1 \end{cases} & & & & \\
 & \xrightarrow{\quad y \quad} & & & \\
 & & b \xleftarrow{\$} \{0,1\} & & \\
 & & c := \begin{cases} y & \text{if } b = 0 \\ y \oplus r_{R^*} & \text{if } b = 1 \end{cases} & & \\
 & & & \xrightarrow{(r_{R^*}, c)} & \\
 & & & \xleftarrow{\quad b' \quad} & \\
 & & d' := \begin{cases} 0 & \text{if } b' = b \\ 1 & \text{if } b' \neq b \end{cases} & & \\
 & \xleftarrow{\quad d' \quad} & & & \\
 & \mathcal{R} \text{ wins if } d' = d & & &
\end{array}
$$

Figure 14: Game $\mathsf{G}_{\mathsf{PRG}}$

Clearly $\mathcal{R}$ is a PPT algorithm since D' is a PPT algorithm. To analyze the win rate of $\mathcal{R}$, observe that, as mentioned earlier, for all $n \in \mathbb{N}$:

- If $d = 0$, meaning $y = G(x)$ for some $x \xleftarrow{\$} \{0,1\}^n$, then the c that $\mathcal{R}$ sends to D' is distributed identically to the c in $\mathsf{G}_{\mathsf{hiding}}$. Consequently, the probability that D' correctly guesses b in $\mathsf{G}_{\mathsf{PRG}}$ is the same as the probability that D' wins $\mathsf{G}_{\mathsf{hiding}}$, that is,

$$\Pr_{b \xleftarrow{\$} \{0,1\}, y \leftarrow G(U_n)} [b' = b \mid d = 0] = \Pr_{b \xleftarrow{\$} \{0,1\}, s \xleftarrow{\$} \{0,1\}^n} \left[D' \text{ wins } \mathsf{G}_{\mathsf{hiding}}\right] = \frac{1}{2} + \frac{1}{2}\varepsilon(n). \tag{6.3}$$

- If $d = 1$, meaning $y \xleftarrow{\$} \{0,1\}^{3n}$, then we see that since y is uniformly random and independent of r_{R^*}, the resulting $c :=$ if $b = 0$ then y else $y \oplus r_{R^*}$ is uniformly random, regardless of the value of b. Thus, the c that D' receives is independent of b, implying that D' can only correctly guess the value of b with probability exactly $\frac{1}{2}$. (Intuitively speaking, the c that D' receives leaks zero information of the bit b.) In other words,

$$\Pr_{b \xleftarrow{\$} \{0,1\}, y \xleftarrow{\$} \{0,1\}^{3n}} [b' = b \mid d = 1] = \Pr_{b \xleftarrow{\$} \{0,1\}, y \xleftarrow{\$} \{0,1\}^{3n}} [D'(r_{R^*}, c) = b \mid d = 1] = \frac{1}{2} \tag{6.4}$$

since $D'(r_{R^*}, c) \perp\!\!\!\perp b$ and $b \in \{0,1\}$ is uniform.

Putting all together, we see that for all $n \in \mathbb{N}$, the probability that $\mathcal{R}$ wins $\mathsf{G}_{\mathrm{PRG}}$ is

$$
\begin{aligned}
&\Pr_{d \xleftarrow{\$} \{0,1\}}\left[\mathcal{R} \text{ wins } \mathsf{G}_{\mathrm{PRG}} : y \leftarrow \begin{Bmatrix} G(U_n) & \text{if } d = 0 \\ U_{3n} & \text{if } d = 1 \end{Bmatrix}\right] \\
=& \Pr_{d \xleftarrow{\$} \{0,1\}}\left[d' = d : y \leftarrow \begin{Bmatrix} G(U_n) & \text{if } d = 0 \\ U_{3n} & \text{if } d = 1 \end{Bmatrix}\right] \\
=& \frac{1}{2} \Pr_{d \xleftarrow{\$} \{0,1\}, y \leftarrow G(U_n)}\left[d' = 0 \,\middle|\, d = 0\right] + \frac{1}{2} \Pr_{d \xleftarrow{\$} \{0,1\}, y \xleftarrow{\$} \{0,1\}^{3n}}\left[d' = 0 \,\middle|\, d = 1\right] \\
=& \frac{1}{2} \Pr_{d \xleftarrow{\$} \{0,1\}, y \leftarrow G(U_n)}\left[b' = b \,\middle|\, d = 0\right] + \frac{1}{2} \Pr_{d \xleftarrow{\$} \{0,1\}, y \xleftarrow{\$} \{0,1\}^{3n}}\left[b' \neq b \,\middle|\, d = 1\right] \\
=& \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\varepsilon(n)\right) + \frac{1}{2}\left(1 - \frac{1}{2}\right) && \text{(by (6.3) and (6.4))} \\
=& \frac{1}{2} + \frac{1}{4}\varepsilon(n),
\end{aligned}
$$

where the advantage $\frac{1}{4}\varepsilon(n)$ is nonnegligible.

Since the PPT distinguisher $\mathcal{R}$ distinguishes between $\{G(U_n)\}_{n\in\mathbb{N}}$ and $\{U_{3n}\}_{n\in\mathbb{N}}$) with nonnegligible advantage for all $n \in \mathbb{N}$, G is not a PRG, contradicting our assumption from the start that G is.

⊛

Answer (c). Let S^* be an unbounded adversarial sender and $n \in \mathbb{N}$. We show that S^* can open the commitment as both $b = 0$ and $b = 1$ with only negligible probability. Given $(\mathsf{out}_{S^*} = (d^0_{S^*}, d^1_{S^*}), \tau = (r, c)) \leftarrow \langle S^*(1^n), R(1^n)\rangle$, we reason

$$
\begin{aligned}
& \mathsf{Open}(0, d^0_{S^*}, \tau = (r, c)) = \mathsf{Open}(1, d^1_{S^*}, \tau = (r, c)) = \top \\
\iff & \mathsf{Com}(d^0_{S^*}, r, 0) = \mathsf{Com}(d^1_{S^*}, r, 1) = c \\
\iff & G(d^0_{S^*}) = G(d^1_{S^*}) \oplus r = c \\
\implies & G(d^0_{S^*}) \oplus G(d^1_{S^*}) = r.
\end{aligned} \tag{6.5}
$$

(Intuitively,

$$\mathsf{Open}(0, d^0_{S^*}, \tau = (r, c)) = \top \wedge \mathsf{Open}(1, d^1_{S^*}, \tau = (r, c)) = \top$$

means that despite the fact that the value of c is fixed after the commit phase, the adversary S^* can still choose at will whether to open c as $b = 0$ or $b = 1$ by sending $d^0_{S^*}$ or $d^1_{S^*}$, respectively, to the honest receiver R in the reveal phase.)

Notice in (6.5) that $d^0_{S^*}, d^1_{S^*} \in \{0,1\}^n$ are forged by the adversarial sender S^* whereas $r \in \{0,1\}^{3n}$ is chosen by the honest receiver R and is thus uniformly random. Even though the distributions of $d^0_{S^*}, d^1_{S^*} \in \{0,1\}^n$ are unknown, observe that there are, however, at most $2^n \cdot 2^n = 2^{2n}$ possible values of $G(d^0_{S^*}) \oplus G(d^1_{S^*})$, far less than all possible values of the uniform $r \in \{0,1\}^{3n}$. Thus, the possibility that the adversarial S^* can find $d^0_{S^*}, d^1_{S^*}$ such that $G(d^0_{S^*}) \oplus G(d^1_{S^*}) = r$ holds is negligible. Formally speaking,

$$\begin{aligned}
\Pr_{r \xleftarrow{\$} \{0,1\}^{3n}} \left[\mathsf{Open}(0, d^0_{S^*}, (r,c)) = \mathsf{Open}(1, d^1_{S^*}, (r,c)) = \top\right] &\leq \Pr_{r \xleftarrow{\$} \{0,1\}^{3n}} \left[G(d^0_{S^*}) \oplus G(d^1_{S^*}) = r\right] \qquad \text{(by (6.5))} \\
&\leq \Pr_{r \xleftarrow{\$} \{0,1\}^{3n}} \left[r \in \mathrm{im}(f)\right] \\
&= \frac{|\mathrm{im}(f)|}{|\{0,1\}^{3n}|} \\
&\leq \frac{|\mathrm{dom}(f)|}{|\{0,1\}^{3n}|} \\
&= \frac{|\{0,1\}^n \times \{0,1\}^n|}{|\{0,1\}^{3n}|} \\
&= \frac{2^{2n}}{2^{3n}} = 2^{-n},
\end{aligned}$$

is negligible, where $f : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^{3n}$ is defined by

$$f(s_1, s_2) \triangleq G(s_1) \oplus G(s_2).$$

Putting all together, we have

$$\begin{aligned}
&\Pr\left[\begin{array}{c} \mathsf{out}_{S^*} = (d^0_{S^*}, d^1_{S^*}) \\ \mathsf{Open}(0, d^0_{S^*}, \tau = (r,c)) = \mathsf{Open}(1, d^1_{S^*}, \tau = (r,c)) = \top \end{array} : (\mathsf{out}_{S^*}, \tau = (r,c)) \leftarrow \langle S^*(1^n), R(1^n)\rangle\right] \\
=& \Pr_{r \xleftarrow{\$} \{0,1\}^{3n}} \left[\mathsf{Open}(0, d^0_{S^*}, (r,c)) = \mathsf{Open}(1, d^1_{S^*}, (r,c)) = \top : ((d^0_{S^*}, d^1_{S^*}), \cdot, (\cdot, c)) \leftarrow \langle S^*(1^n), R(1^n)\rangle\right] \\
\leq& 2^{-n} = \mathsf{negl}(n),
\end{aligned}$$

which, by definition, means that the interactive commitment scheme $(\langle S, R\rangle, \mathsf{Open})$ is statistically binding. ⊛

Answer (d). To show there is no interactive commitment scheme that satisfies both statistical hiding and statistical binding, we instead show that for all interactive commitment scheme, statistical-hiding implies non-statistical-binding.

> Intuition. If an interactive commitment scheme is statistical hiding, then the commitments for 0 and 1 generated by an honest sender are statistically indistinguishable to any unbounded distinguisher. This however allows an unbounded adversarial sender to open either commitment as any bit in the reveal phase albeit the commitment has already been sent to the receiver and cannot be altered.

Let $(\langle S, R\rangle, \mathsf{Open})$ be an interactive commitment scheme and assume that it is statistical hiding. WLOG, suppose that in every execution $\langle S(b, 1^n), R(1^n)\rangle \to (d_S, \tau)$, the decommitment $d_S \in \{0,1\}^{l(n)}$ is always $l(n)$-bit, where $l = O(\mathsf{poly}(n))$. For convenience, let random variables $d^b_{\langle S,R\rangle}$ and $\tau^b_{\langle S,R\rangle}$ denote the decommitment and transcript of the execution $\langle S(b, 1^n), R(1^n)\rangle$ for $b \in \{0,1\}$, i.e.,

$$(d^b_{\langle S,R\rangle}), \tau^b_{\langle S,R\rangle}) \leftarrow \langle S(b, 1^n), R(1^n)\rangle \quad \forall b \in \{0,1\},$$

and let r_R, r_S denote the internal randomness of R, S, respectively. Our goal is to show that $(\langle S, R\rangle, \mathsf{Open})$ is not statistical binding.

> Goal. $(\langle S, R\rangle, \mathsf{Open})$ is not statistical binding, i.e., there exists an unbounded sender S^* such that
>
> $$\Pr\left[\begin{array}{c}\mathsf{out}_{S^*} = (d^0_{S^*}, d^1_{S^*}) \\ \mathsf{Open}(0, d^0_{S^*}, \tau) = \mathsf{Open}(1, d^1_{S^*}, \tau) = \top\end{array} : (\mathsf{out}_{S^*}, \tau) \leftarrow \langle S^*(1^n), R(1^n)\rangle\right]$$
>
> is nonnegligible (as a function of n).

We first derive from the formal guarantee of statistical hiding by taking $R^* \triangleq R$ (the honest receiver) that for every unbounded distinguisher D and all $n \in \mathbb{N}$,

$$\begin{aligned}
&\left|\Pr\left[D\left(\mathsf{view}_R^{\langle S,R^*\rangle}((0,1^n), 1^n; r_S, r_R)\right) = 1\right] - \Pr\left[D\left(\mathsf{view}_R^{\langle S,R\rangle}((1,1^n), 1^n; r_S, r_R)\right) = 1\right]\right| \le \mathsf{negl}(n) \\
\iff &\left|\Pr\left[D\left(1^n, r_R, \tau^0_{\langle S,R\rangle}\right) = 1\right] - \Pr\left[D\left(1^n, r_R, \tau^1_{\langle S,R\rangle}\right) = 1\right]\right| \le \mathsf{negl}(n),
\end{aligned}$$

Since distinguishing between $(1^n, r_R, \tau^0_{\langle S,R\rangle})$ and $(1^n, r_R, \tau^1_{\langle S,R\rangle})$ is no harder than distinguishing between $\tau^0_{\langle S,R\rangle}$ and $\tau^1_{\langle S,R\rangle}$, for every unbounded distinguisher D' and all $n \in \mathbb{N}$, we also have

$$\left|\Pr\left[D'\left(\tau^0_{\langle S,R\rangle}\right) = 1\right] - \Pr\left[D'\left(\tau^1_{\langle S,R\rangle}\right) = 1\right]\right| \le \mathsf{negl}(n). \tag{6.6}$$

(To be exact, this is because given a distinguisher D' of $\tau^0_{\langle S,R\rangle}$ and $\tau^1_{\langle S,R\rangle}$ with advantage $\varepsilon(n)$, we may in fact construct a distinguisher D of $(1^n, r_R, \tau^0_{\langle S,R\rangle})$ and $(1^n, r_R, \tau^1_{\langle S,R\rangle})$ with advantage greater or equal to $\varepsilon(n)$.) Intuitively, since the scheme is statistical hiding, the transcript (or rather, the commitment) leaks zero information about the committed bit $b \in \{0,1\}$, so $\tau^0_{\langle S,R\rangle}$ and $\tau^1_{\langle S,R\rangle}$ must be indistinguishable for any unbounded distinguisher. Since every unbounded distinguisher knows how a commitment is opened (i.e. know Open) and can thus run Open, for any committed bit $b \in \{0,1\}$ and all $n \in \mathbb{N}$, it must be that

$$\Pr\left[\exists d^* \in \{0,1\}^{l(n)} \text{ such that } \mathsf{Open}(\bar{b}, d^*, \tau^b_{\langle S,R\rangle}) = \top\right] = 1 - \mathsf{negl}(n),$$

which means that there exists a decommitment d^* that opens $\tau^b_{\langle S,R\rangle}$ as the other bit $\bar{b}$. We show that if that were not the case, then (6.6) would fail to hold.

Claim 6.1. For any $b \in \{0,1\}$ and all $n \in \mathbb{N}$,

$$\Pr\left[\exists d^* \in \{0,1\}^{l(n)} \text{ such that } \mathsf{Open}(\bar{b}, d^*, \tau^b_{\langle S,R\rangle}) = \top\right] = 1 - \mathsf{negl}(n).$$

Proof. Suppose to the contrary that for some $b \in \{0,1\}$ (WLOG, assume $b = 0$) and all $n \in \mathbb{N}$

$$\Pr\left[\exists d^* \in \{0,1\}^{l(n)} \text{ such that } \mathsf{Open}(\bar{b} = 1, d^*, \tau^{b=0}_{\langle S,R\rangle}) = \top\right] = 1 - \varepsilon(n), \tag{6.7}$$

where $\varepsilon(n)$ is nonnegligible. Then the following unbounded distinguisher D' distinguishes commitments (transcripts) to 0 and 1, namely $\tau^0_{\langle S,R\rangle}$ and $\tau^1_{\langle S,R\rangle}$, with non-negl. prob. for infinitely many $n \in \mathbb{N}$:

> On input $\tau \leftarrow \tau^b_{\langle S,R\rangle}$ for $b \xleftarrow{\$} \{0,1\}$, check whether there exists $d^* \in \{0,1\}^{l(n)}$ such that $\mathsf{Open}(1, d^*, \tau) = \top$. If true, then output $b' = 1$; otherwise output $b' = 0$.

To see that the advantage of D' is nonnegligible, we calculate

$$\begin{aligned}
&\left|\Pr\left[D'\left(\tau^0_{\langle S,R\rangle}\right) = 1\right] - \Pr\left[D'\left(\tau^1_{\langle S,R\rangle}\right) = 1\right]\right| \\
=&\left|\underbrace{\Pr\left[\exists d^* \in \{0,1\}^{l(n)}.\mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top\right]}_{= 1 - \varepsilon(n) \ \text{(by (6.7))}} - \underbrace{\Pr\left[\exists d^* \in \{0,1\}^{l(n)}.\mathsf{Open}(1, d^*, \tau^1_{\langle S,R\rangle}) = \top\right]}_{\substack{\overset{🍄}{\geq} \Pr\left[\mathsf{Open}(1, d^1_{\langle S,R\rangle}, \tau^1_{\langle S,R\rangle}) = \top\right] \\ = 1 \ \text{(by the \textbf{correctness} of } (\langle S,R\rangle, \mathsf{Open}))}}\right| \\
\geq&\left|(1 - \varepsilon(n)) - 1\right| = \varepsilon(n),
\end{aligned}$$

which is nonnegligible, implying the advantage of D' is also nonnegligible. Note that the inequality (🍄) follows from the fact that

$$\mathsf{Open}(1, d^1_{\langle S,R\rangle}, \tau^1_{\langle S,R\rangle}) = \top \implies \exists d^* \in \{0,1\}^{l(n)}.\mathsf{Open}(1, d^*, \tau^1_{\langle S,R\rangle}) = \top,$$

where $d^1_{\langle S,R\rangle}$ is exactly the correct decommitment for committed bit $b = 1$ and transcript $\tau^1_{\langle S,R\rangle}$. However, this contradicts (6.6), which states that D' has only negligible advantage in distinguishing between $\tau^0_{\langle S,R\rangle}$ and $\tau^1_{\langle S,R\rangle}$. ∎

With Claim 6.1, we can construct as follows an unbounded adversarial sender S^* that opens the commitment (transcript) $\tau^0_{\langle S,R\rangle}$ as both $b = 0$ and $b = 1$, thereby breaking the requirement of statistical binding. Roughly speaking, S^* is basically S, with the exception that it outputs as decommitment an additional d^* alongside the default $d^0_{\langle S,R\rangle}$, where $d^* \in \{0,1\}^{l(n)}$ is such that $\mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top$. Note that S^* can find such d^* by enumerating all values in $\{0,1\}^{l(n)}$ since S^* has an unbounded time limit and Claim 6.1 guarantees that such d^* exists with very high probability.

> On input 1^n, run $S(0, 1^n)$ (the honest sender) as a subroutine and, during its execution, forward all messages from R to S and from S to R, while recording all forwarded messages as the transcript $\tau = \tau^0_{\langle S,R\rangle}$. Upon the termination of S, collect its output $d_S = d^0_{\langle S,R\rangle}$. As an extra step, find $d^* \in \{0,1\}^{l(n)}$ such that $\mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top$. Finally, output $(d^0_{\langle S^*,R\rangle}, d^1_{\langle S^*,R\rangle}) := (d^0_{\langle S,R\rangle}, d^*)$ if d^* is found, and abort (i.e. output $(d^0_{\langle S^*,R\rangle}, d^1_{\langle S^*,R\rangle}) := (d^0_{\langle S,R\rangle}, \perp)$) otherwise.

The following diagram illustrates the execution $\langle S^*(1^n), R(1^n)\rangle \to ((d^0_{\langle S^*,R\rangle}, d^1_{\langle S^*,R\rangle}), \tau_{\langle S^*,R\rangle})$.

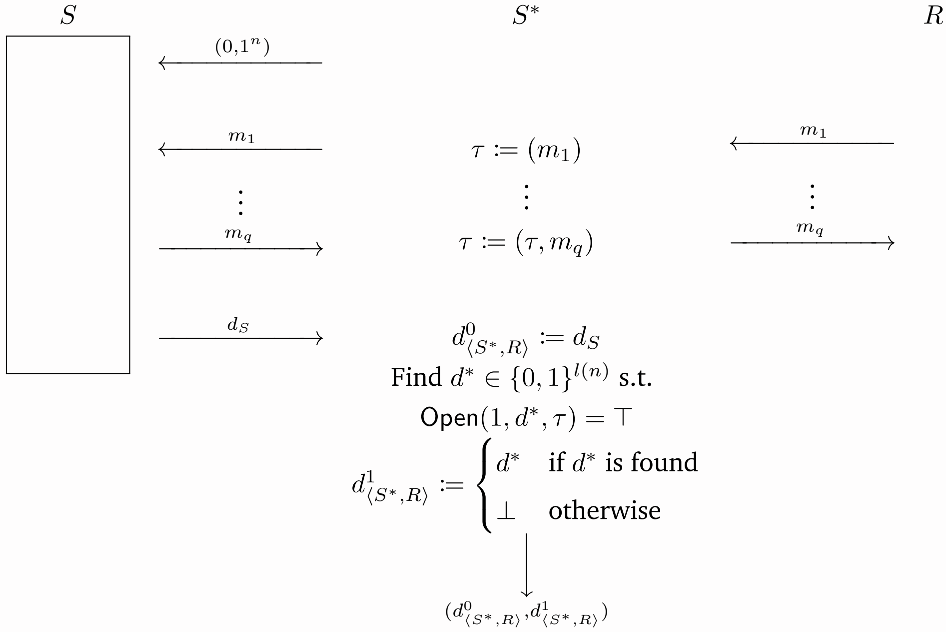

Figure 15: Execution $\langle S^*(1^n), R(1^n)\rangle \to ((d^0_{\langle S^*,R\rangle}, d^1_{\langle S^*,R\rangle}), \tau_{\langle S^*,R\rangle})$

Note that since S^* acts as the man in the middle between S and R,

$$\tau_{\langle S^*,R\rangle} = \tau = \tau^0_{\langle S,R\rangle} \text{ and } d^0_{\langle S^*,R\rangle} = d_S = d^0_{\langle S,R\rangle}. \tag{6.8}$$

For all $n \in \mathbb{N}$, the probability that S^* finds two valid decommitments for $b = 0$ and $b = 1$ is thus

$$\begin{aligned}
&\Pr\left[\begin{array}{c} \mathsf{out}_{S^*} = (d^0_{\langle S^*,R\rangle}, d^1_{\langle S^*,R\rangle}) \\ \mathsf{Open}(0, d^0_{\langle S^*,R\rangle}, \tau_{\langle S^*,R\rangle}) = \mathsf{Open}(1, d^1_{\langle S^*,R\rangle}, \tau_{\langle S^*,R\rangle}) = \top \end{array} : (\mathsf{out}_{S^*}, \tau_{\langle S^*,R\rangle}) \leftarrow \langle S^*(1^n), R(1^n)\rangle\right] \\
&\geq \Pr\Big[\mathsf{Open}(0, d^0_{\langle S^*,R\rangle}, \tau_{\langle S^*,R\rangle}) = \mathsf{Open}(1, d^1_{\langle S^*,R\rangle}, \tau_{\langle S^*,R\rangle}) = \top \wedge d^1_{\langle S^*,R\rangle} \neq \perp \\
&\qquad\qquad : ((d^0_{\langle S^*,R\rangle}, d^1_{\langle S^*,R\rangle}), \tau_{\langle S^*,R\rangle}) \leftarrow \langle S^*(1^n), R(1^n)\rangle\Big] \\
&= \Pr\left[\mathsf{Open}(0, d^0_{\langle S,R\rangle}, \tau^0_{\langle S,R\rangle}) = \mathsf{Open}(1, d^1_{\langle S^*,R\rangle}, \tau^0_{\langle S,R\rangle}) = \top \wedge (\exists d^* \in \{0,1\}^{l(n)}.\mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top)\right] \qquad \text{(by (6.8))} \\
&= \Pr\left[\exists d^* \in \{0,1\}^{l(n)}.\mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top\right] \\
&\qquad \Pr\left[\mathsf{Open}(0, d^0_{\langle S,R\rangle}, \tau^0_{\langle S,R\rangle}) = \mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top \mid \mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top\right] \\
&= \underbrace{\Pr\left[\exists d^* \in \{0,1\}^{l(n)}.\mathsf{Open}(1, d^*, \tau^0_{\langle S,R\rangle}) = \top\right]}_{= 1 - \mathsf{negl}(n) \text{ (by Claim 6.1)}} \cdot \underbrace{\Pr\left[\mathsf{Open}(0, d^0_{\langle S,R\rangle}, \tau^0_{\langle S,R\rangle}) = \top\right]}_{= 1 \text{ (by the \textbf{correctness} of } (\langle S,R\rangle, \mathsf{Open}))} = 1 - \mathsf{negl}(n),
\end{aligned}$$

which is nonnegligible, implying the probability is also nonnegligible. This implies that $(\langle S, R\rangle, \mathsf{Open})$ is not statistical binding, as desired.

Hence, there can't be interactive commitment scheme that is both statistically hiding and statistically binding. ⊛

Reference

P1 (None)

P2
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P3 (None)

P4 (None)

P5 (None)

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